Strong Colorings of Orthogonality Hypergraphs as Noncontextual Eigenvalue Assignments

Karl Svozil [a)]
Institute for Theoretical Physics, TU Wien, Wiedner Hauptstraße 8-10/136, A-1040 Vienna, Austria

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Suppose intertwined complete experiments each measure a maximal observable with one common spectrum, and the eigenvalue of a shared outcome is an element of physical reality, the same in every experiment. Such noncontextual eigenvalues exist exactly when the context hypergraph can be colored with as many colors as the dimension. They can fail even when the two-valued states separate all outcomes, so that the logic is classically realizable; completed Yu–Oh is an example. When no such assignment exists, any classical model needs more hidden states than the dimension, just to tell apart preparations along orthogonal rays. Where noncontextual eigenvalues exist, they can force two distinct rays to share an eigenvalue, a chromatic form of Kochen–Specker inseparability; for one quantum preparation, every classical model must put a third of its weight on truth assignments that no eigenvalue assignment produces. We also correct an earlier inference from coloring to geometry.

I. INTRODUCTION

An experimental context with d rank-one outcomes admits a nondegenerate observable with d distinct eigenvalues. These eigenvalues distinguish all d outcomes and provide the maximal resolution of a complete projective experiment within that context. Such an observable is maximal in the sense that every self-adjoint observable commuting with it is a function of it, so the statistics of the latter can be obtained by processing its measurement outcomes. Noninjective functions merge outcomes and reduce this resolution. This is the finite-dimensional setting of von Neumann's functional representation of commuting observables[1] (Satz 8, p. 221); see also Halmos[2] (§84, pp. 171–172).

When complete experiments are intertwined, they share outcomes represented by the same rank-one projectors. We ask whether the eigenvalues of the maximal observables measured in such intertwined experiments can be assigned to their outcomes noncontextually, in the spirit of the elements of physical reality of Einstein, Podolsky, and Rosen[3]: whether every outcome carries a definite eigenvalue that belongs to the outcome itself, rather than to the experiment in which it is recorded. Three assumptions make this precise.

(i) *Maximality.* Each complete experiment measures a maximal observable, so its d outcomes are never merged and carry d distinct eigenvalues. A degenerate observable would discard part of what the experiment resolves.

(ii) *Uniformity.* All maximal observables share one spectrum of d eigenvalues $\Sigma = \{\lambda_0, \ldots, \lambda_{d-1}\}$. The eigenvalues are nothing but labels of the d outcomes of a complete experiment. Since self-adjoint operators with a common nondegenerate spectrum are unitarily equivalent, uniformity says that every complete experiment measures one and the same quantity in a different orientation, $A_e = U_e A U_e^\dagger$.

(iii) *Noncontextuality of eigenvalues.* An outcome shared by several complete experiments carries the same eigenvalue in each of them, just as Kochen and Specker require a shared outcome to carry the same truth value[4].

[a)] Electronic mail: karl.svozil@tuwien.ac.at

No constraint is placed on how the d eigenvalues are distributed over the outcomes of different contexts; any permutation within a context is allowed. A noncontextual eigenvalue assignment in this sense labels the outcomes, the most elementary record a complete experiment produces, and assumption (iii) treats these labels as elements of physical reality. For spin one, an example is the observable $aS_x^2+bS_y^2+cS_z^2$ with distinct real a, b, c considered by Kochen and Specker[4]: every orthogonal triad of directions yields one such observable, with the same eigenvalues $b+c$, $a+c$, and $a+b$. A noncontextual eigenvalue assignment then gives each direction the same coefficient, a, b, or c, in every triad that contains it.

These assumptions concern eigenvalues rather than which outcome occurs, yet they have immediate consequences for truth assignments. A two-valued state selects one outcome in every context and retains only the distinction between that outcome and the others. A noncontextual eigenvalue assignment yields d such states at once, one for each eigenvalue λ_a: "the outcome with eigenvalue λ_a occurs in every experiment." These d truth assignments have pairwise disjoint supports covering every outcome. A noncontextual eigenvalue assignment is thus a hidden-variable model whose hidden variable is the eigenvalue itself; its d equally weighted values reproduce the maximally mixed state. Because this requirement is more demanding than Kochen–Specker noncontextuality, which asks for a single truth assignment, its failure—*chromatic contextuality*—is a weaker form of contextuality that already appears in configurations retaining many classical truth assignments. Even when a noncontextual eigenvalue assignment exists, its consistency across intertwined contexts can force relations between distinct outcomes, such as equal eigenvalues for nonorthogonal rays, that are absent from the full family of individual truth assignments. We study both the existence of noncontextual eigenvalue assignments and the event distinctions and logical relations that survive when such assignments are available[5–7].

To express these questions, let a d-uniform context hypergraph $H = (V, E)$ encode the complete experiments. An ordinary two-valued state $s : V \to \{0, 1\}$ selects exactly one outcome in every context, so $\sum_{v\in e} s(v) = 1$ for every $e \in E$. The Kochen–Specker (KS) theorem[4] establishes the nonexistence of such a global truth assignment for suitable finite families of intertwined projective contexts in dimension $d \geq 3$.

A strong d-coloring $c : V \to \{0, \ldots, d-1\}$ assigns all d distinct labels bijectively within every context and independently of context at shared vertices. It is the formal counterpart of a noncontextual eigenvalue assignment: $c(i) = a$ means that outcome i carries the eigenvalue λ_a in every context containing it, so assumptions (i)–(iii) hold if and only if $\chi(H) \leq d$. For an injective ray realization $i \mapsto [v_i] \in \mathbb{P}(\mathbb{F}^d)$, with $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, rank-one projectors P_i, and a common nondegenerate real spectrum $\Sigma = \{\lambda_0, \ldots, \lambda_{d-1}\}$, it defines the maximal context observables

$$A_e = \sum_{i\in e} \lambda_{c(i)} P_i. \tag{1}$$

Because the indicator functions $s_a(v) = \mathbf{1}_{\{c(v)=a\}}$ $(a \in \{0, \ldots, d-1\})$ of any strong d-coloring form d two-valued states whose supports partition V, one has the strict one-way implication

$$\mathcal{S}(H) = \varnothing \quad \Longrightarrow \quad \chi(H) > d, \tag{2}$$

whose converse fails even when $\mathcal{S}(H)$ is unital and separates every pair of vertices. The earlier introduction in Ref.[6] inferred absence of global two-valued assignments from $\chi > d$. Equation (2) is the valid implication; completed Yu–Oh refutes the converse. Both the implication and the failure of its converse, for the original 13-ray Yu–Oh set, were noted by Cabello[8]. In terms of eigenvalues, if no noncontextual truth assignment exists, no noncontextual eigenvalue assignment exists either; but noncontextual eigenvalues can be impossible while truth assignments, even a unital separating family of them, survive. The distinct Appendix B inference about geometry is corrected below. We say that H has an *exact palette*, or is *chromatically complete*, when $\chi(H) = d$, with no additional density or geometric assumption. This asymmetry separates chromatic contextuality into two distinct regimes that are unified in this work:

1. *Regime I, common-spectrum chromatic obstructions ($\chi(H) > d$, possibly with $\mathcal{S}(H) \neq \varnothing$):* Here no strong d-coloring exists ($\mathcal{C}_d(H) = \varnothing$), so the outcomes admit no noncontextual assignment of d eigenvalues: every family of context observables with one common spectrum of d eigenvalues must either merge outcomes, and so fail to be maximal, or let a shared outcome change its eigenvalue from context to context; maximal observables with noncontextual eigenvalues need more than d eigenvalues. This happens even in examples in which classical truth assignments survive. A critical foundational clarification—correcting an interpretation in Appendix B of the earlier work with Shekarriz[6]—is that the chromatic obstruction $\chi(H) > d$ is an operator spectral-value obstruction and *does not* by itself obstruct geometric ray realizability in $\mathbb{F}^d$. Indeed, for finite 3-uniform linear hypergraphs of Berge girth five with unital separating state spaces, strong 3-colorability and injective $\mathbb{C}^3$ realizability are logically independent (Proposition 10). Ruling out geometric ray assignments for Greechie's Petersen-derived G_{32}[9] requires a direct Hermitian projection-identity argument (Theorem 11) rather than $\chi(G_{32}) = 4$.

2. *Regime II, exact-palette chromatic separability and logical gadgets ($\chi(H) = d$ with $\mathcal{S}_{\text{ext}}(H) \subsetneq \mathcal{S}(H)$):* In this regime, $\mathcal{C}_d(H) \neq \varnothing$, so noncontextual eigenvalue assignments exist, but some two-valued states $s \in \mathcal{S}(H)$ cannot be completed to a strong d-coloring: they are truth assignments that no noncontextual eigenvalue assignment produces. We characterize the extendable states $\mathcal{S}_{\text{ext}}(H)$ by the residual chromatic condition $\chi(\Gamma(H)[V \setminus \operatorname{supp} s]) \leq d-1$ (Theorem 2), which in dimension $d = 3$ yields an odd-cycle nonextendability certificate. In the specified subhypergraphs—such as the faithful real 17-ray and 21-ray restrictions $H_{17,10}$ and $H_{21,13}$ of completed Yu–Oh[7,10] (Theorem 15) and our faithful 3- and 4-dimensional logical gadgets (Secs. IV C and IV D)—ordinary states separate all vertices while every admissible d-coloring forces nonadjacent rays to share a color ($u \sim_c v$, chromatic inseparability / implicit identity[11]) or obeys purely chromatic true-implies-false (TIFS) or ternary exclusions violated by nonextendable states in $\mathcal{S}(H) \setminus \mathcal{S}_{\text{ext}}(H)$.

These two regimes are governed by a shared structural mechanism. Lemma 3 ($\chi(H) = d$ if and only if H admits $d-1$ pairwise disjoint states) is the global unanchored form of Theorem 2. Likewise, the eight normalized 3-colorings of the four base contexts of completed Yu–Oh H_{YO}^{+} partition into four antipodal pairs that each make one of the four diagonal neighborhoods rainbow (Theorem 13): retaining all four diagonal sectors forces $\chi(H_{\text{YO}}^{+}) = 4$ (Regime I), whereas retaining two or three sectors leaves four or two normalized 3-colorings and forces two or 21 chromatically inseparable pairs in $H_{17,10}$ and $H_{21,13}$ (Regime II), which our exhaustive 65,535-subset inventory classification (Proposition 16) proves minimal among the faithful restrictions of H_{YO}^{+} with inherited coordinates.

For truth values, elements of reality fail as in the Kochen–Specker theorem: the logic has no two-valued states, or too few to separate its outcomes, and then it has no classical realization. For eigenvalues the situation is different. If the two-valued states of H are unital and separating, H can be realized classically as a partition logic[4,12,13]: outcome i becomes the set of states with $s(i) = 1$, and each context becomes a partition of this set. Noncontextual eigenvalues can still be impossible, because they require a strong d-coloring, in the classical realization as much as in the quantum one. Greechie's G_{32} is an example: it has a classical realization, no orthogonal ray assignment in $\mathbb{C}^3$ (Theorem 11), and $\chi(G_{32}) = 4$. Thus $\chi(H) > d$ is a condition on the spectra of maximal observables, not on truth values. It does, however, bound the size of every classical model. Preparations along orthogonal rays can be told apart with certainty, so in any hidden-variable model they have disjoint sets of hidden states; picking one hidden state for each ray then colors $\Gamma(H)$, and the model needs at least $\chi(H)$ hidden states (Sec. VIII works this out for Yu–Oh). A noncontextual eigenvalue assignment keeps orthogonal rays apart with only d hidden states, the eigenvalues themselves. Thus, when H is realized in $\mathbb{C}^d$, $\chi(H) > d$ means that a classical model needs more hidden states than the quantum system has dimensions merely to tell apart preparations along orthogonal rays, a gap that Prakash[14] calls representational contextuality and turns into a

memory advantage of quantum automata. A stronger separation, which excludes every local classical model, arises when an experiment fixes the label of each outcome, as the d-party supersinglet of the next paragraph does.

Noncontextual eigenvalues and strong colorings were first related by Cabello[8]. The first version of that preprint assigned noncontextual values $-1, 0, 1$ to the spin-one components S_u along the rays u and concluded that $\chi > d$ characterizes state-independent contextuality. For an orthogonal triad u, v, w, however, the eigenvectors of S_u, S_v, S_w for the eigenvalue 1 (or -1) are not mutually orthogonal, so quantum mechanics fixes only which of them takes the value 0; the necessity of $\chi > d$ and the minimality of the 13-ray Yu–Oh set were later published in Ref.[15]. The context observables A_e of Eq. (1) are diagonal in their contexts, so assumptions (i)–(iii) are equivalent to $\chi(H) \leq d$ without such a gap. The second version of the preprint distributes the d particles of the totally antisymmetric supersinglet among d parties. If all parties measure bases containing a ray v, exactly one of them obtains v; if $d-1$ parties measure the same basis, the ray of that basis which none of them obtains is obtained with certainty by the last party in any basis containing it. A local hidden-variable model of these perfect correlations exists exactly when the hypergraph of the measured bases has a strong d-coloring, the color of a ray being the party that obtains it; for the Yu–Oh set this hypergraph is H_{YO}^{+}. Here the label of an outcome is the party that records it, and its noncontextuality, assumption (iii), follows from locality and the EPR criterion of reality[3] instead of being postulated. Quantum mechanics produces these correlations whenever H has an orthogonal ray assignment in $\mathbb{C}^d$, while local classical models need a strong d-coloring, so here $\chi(H) > d$ does separate quantum from classical physics. G_{32} has no ray assignment and gives no such separation. For a single system in the maximally mixed state the threshold is fractional: I_d/d has a noncontextual truth-assignment model exactly when $\chi_f(\Gamma(H)) = d$[16] (Proposition 6). Contrary to the claim of Ref.[8], $\chi > d$ is therefore not sufficient for the contextuality of I_d/d: the faithful complex Hesse hypergraph H_{37} has $\chi = 4$ and $\chi_f = 3$ (Theorem 19). The gap between χ and the least dimension of an orthogonal ray assignment has also been studied in connection with the quantum chromatic number[17].

We further connect these combinatorial and geometric layers to orthomodular event representations and quantum probability polytopes $\mathcal{P}_{\mathrm{ext}}(H) \subseteq \mathcal{P}(H) = \operatorname{conv} \mathcal{S}(H)$, where $\mathcal{P}_{\mathrm{ext}}(H)$ collects the probability vectors of hidden-variable models that choose a noncontextual eigenvalue assignment and one of its d eigenvalues. On $H_{17,10}$, an explicit quantum ray preparation lies in $\mathcal{P}(H_{17,10}) \setminus \mathcal{P}_{\mathrm{ext}}(H_{17,10})$ and requires nonextendable state weight exactly $1/9$; the preparation along the coordinate ray z_3 requires exactly $1/3$, the largest weight that a single chromatic identity of $H_{17,10}$ can certify (Theorem 24). In complex dimension three, the 37-ray Hesse restriction H_{37}[18] has no injective $\mathbb{R}^3$ realization, pastes into a 76-event orthomodular lattice with exact minimum state-family sizes 6, 7, and 18 for vertex separation, event separation, and order determination (Theorem 19), and demonstrates that exact event-order determination does not guarantee classical mixtures for all Born preparations (Corollary 25). In real dimension four, our two-plane completion lemma (Lemma 21) produces the canonical linear 65-ray hypergraph H_{65} completing the 17-ray core of Haynes *et al.*[19] with $\chi = 5$, $\chi_f = 17/4$, and 393,216 states that fail atom-order determination and collapse a ray with a plane (Theorem 22). Finally, we note that the existence form of the KS ancestry hypothesis follows from an elementary union argument (Remark 26), give an abstract transverse chromatic-to-stateless construction (Proposition 27), establish exact inclusions into specified KS parents (Bub's 49-ray set[20], the 165-ray MUB inventory, Peres's 24-ray geometry, and 68-ray extensions containing the 18-ray parity proof $C_{18,9}$ of Cabello *et al.*[21] and a copy $C_{20,11}$ of Kernaghan's 20-ray parity proof[22]), and contrast abstract context states $\mathcal{S}_{\mathrm{ctx}}$ with geometric states $\mathcal{S}_{\mathrm{geom}}$ on the 36-ray completion and 45-ray vertex deletion of $C_{18,9}$. Complete vector, state, and coloring enumerations and finite verification certificates are provided in the supplementary material, which additionally cites Refs.[23–25].

II. UNIFIED FRAMEWORK: STATES, COLORINGS, SEPARABILITY, AND COMPLETION

A. Context hypergraphs, orthogonal realizations, and spectral regimes

All hypergraphs and graphs considered here are finite. A d-uniform *context hypergraph* $H = (V, E)$ ($d \geq 2$) consists of a nonempty collection E of d-element subsets of V, called *contexts*, with $V = \bigcup_{e \in E} e$. Its *two-section* (or *context*) *graph* $\Gamma(H)$ has vertex set V and edge set $E(\Gamma(H)) = \bigcup_{e \in E} \binom{e}{2}$. A hypergraph H is *linear* if $|e \cap e'| \leq 1$ for all distinct $e, e' \in E$. A *Berge cycle* of length $\ell \geq 2$ is an alternating sequence of distinct vertices $x_1, \ldots, x_\ell$ and distinct contexts $e_1, \ldots, e_\ell$ with $\{x_j, x_{j+1}\} \subseteq e_j$ cyclically; the least such ℓ is the *Berge girth* $g(H)$ (or ∞ if acyclic). In Greechie's pasting of Boolean blocks[9,26], linearity ($g(H) \geq 3$) with $g(H) \geq 4$ yields an *orthomodular poset*, and $g(H) \geq 5$ yields an *orthomodular lattice* (OML).

With the Hermitian inner product $\langle x, y \rangle = \sum_j \overline{x_j} y_j$ on $\mathbb{F}^d$ ($\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$), we distinguish three levels of geometric realization:

1. An *orthogonal ray assignment* $\rho : V \to \mathbb{P}(\mathbb{F}^d)$, $i \mapsto [v_i]$ ($v_i \neq 0$), requires $\langle v_i, v_j \rangle = 0$ whenever $i \sim j$ in $\Gamma(H)$; nonadjacent vertices may share a ray.
2. An *injective orthogonal representation* additionally requires $[v_i] \neq [v_j]$ for all $i \neq j$.
3. An injective representation is *edge-faithful* (abbreviated FOR) if $\langle v_i, v_j \rangle = 0 \iff i \sim j$ in $\Gamma(H)$ for all $i \neq j$[27].

A *strong k-coloring* of H is a map $c : V \to \{0, \ldots, k-1\}$ injective on every context $e \in E$, i.e., a proper k-coloring of $\Gamma(H)$; the least such k is $\chi(H) = \chi(\Gamma(H))$. We call H *chromatically complete* (equivalently, H has an exact palette) when $\chi(H) = d$ and write $\mathcal{C}_d(H)$ for the set of strong d-colorings of H. For a realized d-uniform hypergraph with rank-one projectors $P_i = v_i v_i^\dagger / \langle v_i, v_i \rangle$, three spectral labelings arise: (a) *context-dependent common spectrum*, where bijections $\ell_e : e \to \Sigma = \{\lambda_0, \ldots, \lambda_{d-1}\}$ define $A_e = \sum_{i \in e} \ell_e(i) P_i$ with shared projectors carrying context-dependent eigenvalues; (b) *context-independent vertex labeling* via a strong k-coloring c and distinct real scalars $\lambda_0, \ldots, \lambda_{k-1}$ in Eq. (1), where spectra vary across contexts whenever $|c(V)| > d$ (unavoidable when $\chi(H) > d$); and (c) *chromatic completeness* ($k = d$), which provides both context-independent eigenvalues and a single d-element spectrum Σ for all A_e. Only labeling (c) meets all three assumptions of the Introduction: labeling (a) gives up noncontextual eigenvalues, and labeling (b) keeps them at the price of more than d eigenvalues, so that different contexts have different spectra.

B. Two-valued states, color extendability, and the separation hierarchy

A *two-valued context state* is a function $s : V \to \{0, 1\}$ satisfying

$$\sum_{v \in e} s(v) = 1 \qquad (e \in E). \tag{3}$$

We write $\mathcal{S}(H) = \mathcal{S}_{\mathrm{ctx}}(H)$ for the set of context states. For a ray realization ρ, the *geometric state space*

$$\mathcal{S}_{\mathrm{geom}}(H, \rho) = \{s \in \mathcal{S}(H) : s(i) + s(j) \leq 1 \text{ whenever } i \neq j,\ v_i \perp v_j\} \tag{4}$$

additionally enforces orthogonality exclusions on uncovered pairs[28]; $\mathcal{S}_{\mathrm{geom}}(H, \rho) = \mathcal{S}(H)$ when ρ is edge-faithful. A state family $\Omega \subseteq \mathcal{S}(H)$ is *unital* if every $v \in V$ has $s(v) = 1$ for some $s \in \Omega$, and *vertex separating* if distinct $u, v \in V$ have $s(u) \neq s(v)$ for some $s \in \Omega$, inducing the partition set representation $A_i = \{s \in \Omega : s(i) = 1\}$. On an event poset $(L, \leq)$, Ω is *event separating* if distinct $x, y \in L$ have distinct profiles on Ω, and *order determining* if $x \leq y \iff (\forall s \in \Omega,\ s(x) \leq s(y))$.

Definition 1 (Chromatic separability and extendable states)**.** For chromatically complete H ($\mathcal{C}_d(H) \neq \varnothing$), vertices $u, v \in V$ are *chromatically inseparable*, written $u \sim_c v$, if

$$c(u) = c(v) \qquad \text{for every } c \in \mathcal{C}_d(H), \tag{5}$$

an *implicit identity* in graph coloring[11] (Def. 3.8). We call H *chromatically separating* if $u \sim_c v \implies u = v$. A state $s \in \mathcal{S}(H)$ is *extendable*, written $s \in \mathcal{S}_{\text{ext}}(H)$, if $\operatorname{supp} s = \{v \in V : s(v) = 1\}$ is a color class of some $c \in \mathcal{C}_d(H)$[5–7].

Physically, $u \sim_c v$ means that every noncontextual eigenvalue assignment gives u and v the same eigenvalue, and an extendable state is exactly a truth assignment of the form "the outcome with eigenvalue λ_a occurs" for some noncontextual eigenvalue assignment c and eigenvalue λ_a.

Theorem 2 (Completion criterion)**.** *For $s \in \mathcal{S}(H)$ with $S = \operatorname{supp} s$, the following are equivalent:*

1. *$s \in \mathcal{S}_{\text{ext}}(H)$;*
2. *$\chi(\Gamma(H)[V \setminus S]) \leq d - 1$ (so S is a critical independent set[11] (Def. 3.3));*
3. *s belongs to d states in $\mathcal{S}(H)$ with pairwise disjoint supports partitioning V;*
4. *s belongs to $d - 1$ states in $\mathcal{S}(H)$ with pairwise disjoint supports.*

For $d = 3$, $s \notin \mathcal{S}_{\text{ext}}(H)$ if and only if $\Gamma(H)[V \setminus S]$ contains an odd cycle.

Proof. A strong d-coloring with color class S leaves a $(d-1)$-coloring on $V \setminus S$. Conversely, any $(d-1)$-coloring of $\Gamma(H)[V \setminus S]$ combined with a dth color on S is a strong d-coloring because each context meets S once and forms a $(d-1)$-clique in $V \setminus S$; its color indicators yield (3). Conversely, d states with disjoint supports $S_1, \ldots, S_d$ partitioning V define the strong d-coloring $c(v) = k$ for $v \in S_k$, since each context meets each S_k exactly once; hence (3) $\implies$ (1). Trivially (3) $\implies$ (4), and given $d-1$ disjoint state supports $S_1, \ldots, S_{d-1}$, the complement $s_d(v) = 1 - \sum_{k=1}^{d-1} s_k(v)$ is binary and sums to $d - (d-1) = 1$ on every context, giving (3). For $d = 3$, $\chi(\Gamma(H)[V \setminus S]) \leq 2$ holds if and only if $\Gamma(H)[V \setminus S]$ is bipartite (odd-cycle free). □

Applying the equivalence (3) $\iff$ (4) of Theorem 2 globally without anchoring a state s yields the characterization of chromatic completeness:

Lemma 3 (Disjoint states and strong coloring)**.** *A finite d-uniform hypergraph H satisfies $\chi(H) = d$ if and only if $\mathcal{S}(H)$ contains $d - 1$ states with pairwise disjoint supports.*

Corollary 4 (The separation hierarchy)**.** *For chromatically complete H and $u, v \in V$,*

$$u \sim_c v \quad \iff \quad s(u) = s(v) \text{ for all } s \in \mathcal{S}_{\text{ext}}(H). \tag{6}$$

Thus inseparability by $\mathcal{S}(H)$ implies chromatic inseparability ($\sim_c$), equivalently chromatic separation implies separation by $\mathcal{S}(H)$, whereas the converse fails even for faithful real ray configurations. By Theorem 0 of Kochen and Specker[4], separation of all outcomes by $\mathcal{S}(H)$ is what allows an embedding into a Boolean algebra; chromatic inseparability is the same failure of separation, with $\mathcal{S}(H)$ replaced by the states that noncontextual eigenvalue assignments produce. Moreover, $\mathcal{S}_{\text{ext}}(H)$ is always unital whenever $\mathcal{C}_d(H) \neq \varnothing$.

Proof. The indicator states $s_a(v) = \mathbf{1}_{\{c(v)=a\}}$ over all $c \in \mathcal{C}_d(H)$ and $a \in \{0, \ldots, d-1\}$ are precisely $\mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$. For each fixed coloring, the d supports partition V. The inclusion need not be strict: for a single d-element context every state is extendable. Strict inclusion is an additional property established for the specified examples. □

Proposition 5 (Canonical chromatic quotient and closure test)**.** *Let H be chromatically complete.*

1. *The relation $\sim_c$ is an equivalence relation with no two equivalent vertices in a common context. The quotient $\overline{H} = (V/\!\sim_c, \overline{E})$ is d-uniform and chromatically separating, and pullback along $V \to V/\!\sim_c$ induces a bijection $\mathcal{C}_d(\overline{H}) \to \mathcal{C}_d(H)$.*

2. *For $u \neq v$, adjoining one context $\{u, v, w_1, \ldots, w_{d-2}\}$ with fresh vertices w_i yields chromatic number $d+1$ if and only if $u \sim_c v$, and d otherwise. Since recoloring one vertex with a new color separates any pair, the* separable chromatic number χ_{sep} *of a finite graph, the least t for which some set of proper colorings with at most t colors separates every pair of distinct vertices*[5]*, satisfies $\chi_{\mathrm{sep}} \in \{\chi, \chi+1\}$.*

Proof. (1) Adjacent vertices receive distinct colors, so each context projects injectively to $V/\!\sim_c$, and every $c \in \mathcal{C}_d(H)$ factors uniquely through $V/\!\sim_c$. (2) A coloring $c \in \mathcal{C}_d(H)$ extends to the fresh context if and only if $c(u) \neq c(v)$; if $u \sim_c v$, recoloring u with a $(d+1)$th color separates (u, v) while preserving all old context inequalities. □

The quotient preserves colorings, not an injective realization. Closing an inseparable pair is likewise abstract: faithful coordinates make the pair nonorthogonal, so the added context cannot retain those rays.

Put $\mathcal{P}_{\mathrm{ext}} = \operatorname{conv} \mathcal{S}_{\mathrm{ext}}$ and $\mathcal{P} = \operatorname{conv} \mathcal{S}$. Thus $\mathcal{P}$ is the set of probability vectors of noncontextual truth-assignment models, and $\mathcal{P}_{\mathrm{ext}}$ is the subset produced by models that choose a noncontextual eigenvalue assignment c and an eigenvalue λ_a at random and report, in every context, the outcome with eigenvalue λ_a. A binary vector cannot be a convex combination of distinct binary vectors, since each positive-weight term must agree at every coordinate. Consequently,

$$\mathcal{P}_{\mathrm{ext}} = \mathcal{P} \quad \Longleftrightarrow \quad \mathcal{S}_{\mathrm{ext}} = \mathcal{S}. \tag{7}$$

This is stronger than equality of vertex-separation kernels: the pentagon has separating colors yet a nonextendable state. Pair identities alone do not describe all completion constraints.

C. Fractional coloring and fresh-vertex context completion

Write $\alpha(G)$ for the independence number and $\chi_f(G)$ for the fractional chromatic number of $G = \Gamma(H)$.

Proposition 6 (Fractional threshold and uniform state mixture)**.** *For any finite d-uniform context hypergraph H,*

$$\chi_f(\Gamma(H)) = d \quad \Longleftrightarrow \quad \frac{1}{d}\mathbf{1} \in \mathcal{P}(H) = \operatorname{conv} \mathcal{S}(H). \tag{8}$$

Proof. A state mixture $\sum_{s \in \mathcal{S}(H)} \mu_s s = \frac{1}{d}\mathbf{1}$ assigns weight $d\mu_s$ to the independent set $\operatorname{supp} s$, covering each vertex with weight 1 at total weight d, so $\chi_f(\Gamma(H)) \leq d$; since every context is a d-clique, $\chi_f(\Gamma(H)) \geq \omega(\Gamma(H)) \geq d$. Conversely, a fractional coloring $\{w_I\}$ of total weight d obeys $d \leq \sum_{v \in e} \sum_{I \ni v} w_I = \sum_I w_I |I \cap e| \leq \sum_I w_I = d$ for each $e \in E$, forcing $|I \cap e| = 1$ whenever $w_I > 0$, so $\{w_I/d\}$ is a state mixture equal to $\frac{1}{d}\mathbf{1}$. □

Equation (8) restates, for context states, the fractional-chromatic criterion of Ramanathan and Horodecki[16] for the contextuality of the maximally mixed state.

Proposition 7 (Fresh-vertex context completion)**.** *Let G be a finite core graph with designated d-clique base contexts ($d \geq 3$). Complete selected uncovered edges $\{u, v\}$ into d-cliques using $d-2$ degree-one fresh vertices per edge, covering all core vertices and edges. For the resulting hypergraph H and $G' = \Gamma(H)$,*

$$\begin{aligned} \chi(G') &= \max\{d, \chi(G)\}, \\ \chi_f(G') &= \max\{d, \chi_f(G)\}, \\ |\mathcal{S}(H)| &= \sum_S (d-2)^{m(S)}, \end{aligned} \tag{9}$$

TABLE I. Principal configurations. FOR means injective and edge-faithful; g is Berge girth. Counts refer to all context states and all color-class states.

Configuration	d	$\lvert V\rvert/\lvert E\rvert$	g	χ	χ_f	$\lvert\mathcal{S}\rvert$	$\lvert\mathcal{S}_{\text{ext}}\rvert$	Geometry and principal distinction
One triad	3	3/1	∞	3	3	3	3	Real FOR; colors separate.
$H_{6,3}$	3	6/3	3	3	3	4	3	No injective complex realization.
P_5	3	10/5	5	3	3	11	10	Real FOR; colors separate; ternary exclusion.
H_8	3	14/8	5	3	3	11	7	Repeated-ray assignments; no complex injection.
H_9	3	15/9	5	4	3	8	0	No complex ray assignment; context-critical.
G_{32}	3	15/10	5	4	3	6	0	No complex ray assignment.
$H_{17,10}$	3	17/10	5	3	3	18	10	Real FOR; two color-identical pairs.
$H_{21,13}$	3	21/13	5	3	3	21	6	Real FOR; 21 color-identical pairs.
Completed Yu–Oh	3	25/16	5	4	35/11	24	0	Real FOR; context-critical.
H_{37}	3	37/24	5	4	3	153	0	Complex FOR; no real injection; OML minima 6/7/18.
$H_{\text{YO}}^{(4)}$	4	26/16	2	5	46/11	25	0	Real FOR; common-apex suspension, nonlinear.
$C_{36,18}$	4	36/18	3	5	9/2	0	0	Faithful linear KS completion.
H_{65}	4	65/29	3	5	17/4	393,216	0	Real FOR; false atom order and compound collapse.

where S ranges over independent core supports meeting each base context once and $m(S)$ counts completed edges missed by S.

Proof. Any k-coloring ($k \geq d$) or measurable fractional palette of size $k = \max\{d, \chi_f(G)\}$ assigns disjoint colors/sets of measure 1 to u and v, leaving $k-2 \geq d-2$ colors (or measure $k-2 \geq d-2$) for the $d-2$ fresh vertices. A core support S hitting $\{u, v\}$ forces its fresh vertices to 0, whereas a missed edge allows $d-2$ choices of a single fresh vertex set to 1. □

Table I summarizes the principal configurations studied below.

The logical situation has two distinct branches. If a hypergraph has no admissible two-valued context state, it cannot admit a strong d-coloring, because the color classes of such a coloring would supply d admissible states. The converse implication fails: a noncontextual assignment of d eigenvalues to the outcomes may be impossible while ordinary states, including a unital separating family, remain available.

When a noncontextual eigenvalue assignment does exist, its color classes provide a nonempty unital family of extendable states inside the full context-state space. A particular ordinary state belongs to this restricted family precisely when deleting its support leaves a $(d-1)$-colorable graph. Chromatic identities describe pairs having equal eigenvalues in every noncontextual eigenvalue assignment, and hence equal values in every state of that restricted family; they need not hold throughout the ordinary state space. These assignment questions have to be distinguished from the separate requirements of an injective geometric realization, a representation of event order, and a classical mixture reproducing a specified Born probability vector.

III. INDEPENDENCE OF COLORING AND COORDINATIZATION, PETERSEN OBSTRUCTIONS, AND COMPLETED YU–OH

A. Small abstract examples and four-quadrant independence

The non-linear hypergraph $H_{4,2}$ with contexts $\{1,2,3\}$ and $\{1,2,4\}$ has three states $\{1\}$, $\{2\}$, $\{3,4\}$ (all extendable), forces $3 \sim_c 4$ and $s(3) = s(4)$, and has no injective $\mathbb{C}^3$ realization because two orthogonal rays determine a unique orthogonal-complement ray.

Proposition 8 (Smallest linear chromatically inseparable hypergraph)**.** *The linear 3-uniform hypergraph $H_{6,3}$ with contexts $\{1,2,3\}$, $\{3,4,5\}$, $\{5,6,1\}$ of Berge girth three has four unital separating states $\{1,4\}$, $\{2,5\}$, $\{3,6\}$, $\{2,4,6\}$, a unique strong 3-coloring up to permutation with inseparable classes $\{1,4\}$, $\{2,5\}$, $\{3,6\}$, and three extendable states (as $V \setminus \{2,4,6\} = \{1,3,5\}$ is a triangle). No linear 3-uniform hypergraph with $|V| < 6$ or $|E| < 3$ has a chromatically inseparable pair, and $H_{6,3}$ admits no injective realization in $\mathbb{C}^3$.*

Proof. The triangle $\{1,3,5\}$ in $\Gamma(H_{6,3})$ forces three distinct colors on $1,3,5$, whence $c(4) = c(1)$, $c(2) = c(5)$, and $c(6) = c(3)$. Similarly, in $\mathbb{C}^3$, $\{v_1, v_3, v_5\}$ forms an orthonormal basis, forcing $[v_4] = [v_1]$, $[v_2] = [v_5]$, and $[v_6] = [v_3]$. With $|E| \leq 2$, independent permutations of unshared vertices separate all pairs; by inclusion–exclusion, three linear 3-contexts require $|V| \geq 6$. □

Proposition 9 (Clique number at Berge girth at least four)**.** *If $H = (V, E)$ is an n-uniform linear hypergraph ($n \geq 3$) of Berge girth $g(H) \geq 4$, every clique $K \subseteq V$ of size $|K| \geq 3$ in $\Gamma(H)$ is contained in a unique context $e \in E$. Hence $\omega(\Gamma(H)) = n$, and chromatic incompleteness ($\chi(H) > n$) is equivalent to the chromatic–clique gap $\chi(\Gamma(H)) > \omega(\Gamma(H))$.*

Proof. For any three vertices $u, v, w \in K$, pairwise adjacency gives contexts $e_1 \supseteq \{u, v\}$, $e_2 \supseteq \{v, w\}$, $e_3 \supseteq \{w, u\}$. Distinct e_1, e_2, e_3 would form a Berge 3-cycle, contradicting $g(H) \geq 4$. By linearity, the shared context e containing $\{u, v, w\}$ is unique for $\{u, v\}$ and thus contains all of K. □

Proposition 10 (Logical independence of coloring and coordinatization)**.** *For finite 3-uniform context hypergraphs, all four combinations of chromatic completeness ($\chi(H) = 3$) and injective coordinatizability in $\mathbb{C}^3$ occur even within the orthomodular-lattice admissible class of linear hypergraphs of Berge girth $g(H) \geq 5$[9,26] with unital separating state spaces:*

1. *$\chi = 3$, injectively realizable in $\mathbb{R}^3$: a single context $C_1 = \{\{1,2,3\}\}$ (or P_5, $H_{17,10}$, $H_{21,13}$);*
2. *$\chi = 3$, not injectively realizable in $\mathbb{C}^3$: the 8-context Petersen restriction H_8 (Theorem 11);*
3. *$\chi = 4$, edge-faithfully realizable in $\mathbb{R}^3$: completed Yu–Oh H^+_{YO} (Theorem 13);*
4. *$\chi = 4$, admitting no orthogonal ray assignment in $\mathbb{C}^3$: Greechie's G_{32} and H_9 (Theorem 11).*

Injectivity is essential to quadrant (2): any strong n-coloring c yields a noninjective ray assignment $i \mapsto [e_{c(i)+1}] \in \mathbb{P}(\mathbb{F}^n)$. At Berge girth four (an orthomodular poset that is not a lattice), $E_{\square} = \{\{1,2,3\}, \{3,4,5\}, \{5,6,7\}, \{7,8,1\}\}$ also satisfies $\chi = 3$ while forcing $[v_3] = [v_7] \perp \mathrm{span}(v_1, v_5)$ whenever $[v_1] \neq [v_5]$.

B. Greechie's G_{32}, multipole forcing, and correction of the JMP spectral argument

Greechie's G_{32}[9] (Fig. 6) has 15 vertices and ten contexts

$$\begin{aligned} E(G_{32}) = \{&123, 345, 567, 789, 9\,10\,11, \\ &11\,12\,1, 4\,10\,13, 6\,12\,14, \\ &8\,2\,15, 13\,14\,15\}. \end{aligned} \tag{10}$$

Each vertex lies in two contexts, so G_{32} is dual to the Petersen graph P, $\Gamma(G_{32}) = L(P)$, $\mathrm{Aut}(G_{32}) \cong S_5$, and $\chi(G_{32}) = \chi'(P) = 4$. Its six two-valued state supports satisfy

$$|\mathcal{S}(G_{32})| = 6, \qquad |S_\alpha| = 5, \qquad |S_\alpha \cap S_\beta| = 1 \ (\alpha \neq \beta). \tag{11}$$

These supports yield the $\binom{6}{2} = 15$ two-element vertex profiles $B_i = \{\alpha : i \in S_\alpha\}$. Their nonempty pairwise intersections exclude even two disjoint states, confirming $\chi(G_{32}) = 4$ by Lemma 3.

In Appendix B of Shekarriz and Svozil[6], the chromatic obstruction $\chi(G_{32}) = 4 > 3$ was invoked to argue against a 3-dimensional realization of G_{32}. As Proposition 10 demonstrates via H^+_{YO} ($\chi = 4$, yet faithfully realizable in $\mathbb{R}^3$), $\chi(H) > 3$ rules out only a context-independent 3-value spectral assignment—a noncontextual assignment of the eigenvalues of uniform maximal observables to their outcomes—*not* geometric rays. A genuine geometric exclusion requires the projection identity: for an orthonormal basis $\{x, y, z\} \subset \mathbb{C}^3$ with $u \perp x$ and $w \perp z$,

$$\langle u, w\rangle = \langle u, y\rangle\langle y, w\rangle. \tag{12}$$

Theorem 11 (Multipole forcing and minimal complex obstructions)**.** *Let* $H_8 = G_{32} \setminus \{\{8, 2, 15\}, \{13, 14, 15\}\}$ *(*14 *vertices,* 8 *contexts) and* $H_9 = G_{32} \setminus \{\{13, 14, 15\}\}$ *(*15 *vertices,* 9 *contexts).*

1. *In every orthogonal ray assignment of* H_8 *in* $\mathbb{C}^3$ *(allowing repeated rays), the terminal rays satisfy*

$$[v_2] = [v_8] \qquad \textit{and} \qquad [v_{13}] = [v_{14}], \tag{13}$$

and either $[v_1] = [v_4] = [v_9]$ *or* $[v_3] = [v_7] = [v_{12}]$*. Thus* H_8 *has no injective realization in* $\mathbb{C}^3$*.*

2. H_8 *is chromatically complete (*$\chi(H_8) = 3$*) with* 11 *unital separating states, of which* 7 *are extendable (the* 4 *nonextendable states leave odd cycles in* $\Gamma(H_8)$*; see Table S38). Its* 3 *normalized* 3*-colorings (fixing* $(c(1), c(2), c(3)) = (0, 1, 2)$*) are listed in Table S39, forcing the* 6 *chromatically inseparable pairs* $(1, 9)$*,* $(2, 8)$*,* $(3, 7)$*,* $(4, 6)$*,* $(10, 12)$*,* $(13, 14)$*; deleting any single context of* H_8 *removes all forced color identities (leaving* $6, 20, 20, 6, 20, 20, 6, 6$ *normalized colorings on the remaining active vertices, up to global color permutation). Isolated vertices are dropped: retaining the isolated vertex created by deleting* $\{1, 2, 3\}$*,* $\{7, 8, 9\}$*,* $\{4, 10, 13\}$*, or* $\{6, 12, 14\}$ *multiplies the corresponding count by three, from* 6 *to* 18*.*

3. *Neither* H_9*,* G_{32}*, nor the* 12*-vertex,* 21*-edge subgraph* Γ_0 *of* $\Gamma(H_9)$ *on* $\{1, \ldots, 12\}$ *admits any orthogonal ray assignment in* $\mathbb{C}^3$*. Moreover,* H_9 *is context-minimal and* Γ_0 *is edge-minimal: deleting any context of* H_9 *or edge of* Γ_0 *yields a* 3*-colorable structure realizable in* $\mathbb{R}^3$ *by standard basis rays.*

Proof. (1) Let $v_1, \ldots, v_{14} \in \mathbb{C}^3$ be unit vectors for H_8. In $\{v_1, v_2, v_3\}$, $v_{11}, v_{12} \perp v_1$ and $v_4, v_5 \perp v_3$, so Eq. (12) gives $\langle x, y\rangle = \langle x, v_2\rangle\langle v_2, y\rangle$ for $x \in \{v_{11}, v_{12}\}, y \in \{v_4, v_5\}$. Expanding v_{12} in $\{v_5, v_6, v_7\}$ ($v_{12} \perp v_6$) and v_4 in $\{v_9, v_{10}, v_{11}\}$ ($v_4 \perp v_{10}$) yields $v_7\langle v_7, v_{12}\rangle = v_{12} - v_5\langle v_5, v_{12}\rangle$ and $v_9\langle v_9, v_4\rangle = v_4 - v_{11}\langle v_{11}, v_4\rangle$. Since $v_7 \perp v_9$, $v_{12} \perp v_{11}$, and $v_5 \perp v_4$, their inner product vanishes:

$$\begin{aligned} 0 &= \langle v_{12}, v_4\rangle + \langle v_{12}, v_5\rangle\langle v_5, v_{11}\rangle\langle v_{11}, v_4\rangle \\ &= \langle v_{12}, v_2\rangle\langle v_2, v_4\rangle\big(1 + |\langle v_2, v_5\rangle|^2|\langle v_2, v_{11}\rangle|^2\big). \end{aligned} \tag{14}$$

Thus $\langle v_2, v_4\rangle = 0$ or $\langle v_{12}, v_2\rangle = 0$. If $\langle v_2, v_4\rangle = 0$, set $(v_1, v_3, v_2) = (e_1, e_2, e_3)$; then $v_4 \perp v_2, v_3 \implies [v_4] = [e_1]$, $\{3, 4, 5\} \implies [v_5] = [e_3] = [v_2]$, $\{9, 10, 11\}$ with $v_{10}, v_{11} \perp e_1$ forces $[v_9] = [e_1]$, $\{5, 6, 7\}$ and $\{7, 8, 9\}$ force $[v_7] = [e_2], [v_6] = [e_1], [v_8] = [e_3] = [v_2]$, and $\{4, 10, 13\}$, $\{9, 10, 11\}$, $\{11, 12, 1\}$, $\{6, 12, 14\}$ in $\mathrm{span}(e_2, e_3)$ force $[v_{13}] = [v_{11}] = [v_{14}]$ and $[v_1] = [v_4] = [v_9]$. Reflection symmetry gives the second branch when $\langle v_{12}, v_2\rangle = 0$.

(2) In H_8, fixing $(c(1), c(2), c(3)) = (0, 1, 2)$ leaves 16 choices along $345, 567, 789, 9\,10\,11$, of which 13 fail on $11\,12\,1, 4\,10\,13, 6\,12\,14$, leaving c_1, c_2, c_3 in Table S39. Equal columns identify the 6 pairs.

(3) Both H_9 (hence G_{32}) and Γ_0 add the orthogonality $v_2 \perp v_8$ to H_8, contradicting $[v_2] = [v_8]$. For minimality, deleting $\{8, 2, 15\}$ from H_9 leaves H_8 ($\chi = 3$); deleting $\{4, 10, 13\}$ or $\{6, 12, 14\}$ is isomorphic by D_6 symmetry to deleting $\{8, 2, 15\}$; and deleting any ring context (say $\{1, 2, 3\}$) allows the 3-coloring $(c(1), \ldots, c(15)) = (2, 1, 2, 0, 1, 0, 2, 0, 1, 2, 0, 1, 1, 2, 2)$. On Γ_0, the following deletions cover all 21 edges up to D_6 symmetry: $\{1, 2\}$ (colored by 001020120121), $\{2, 3\}$ (011020120121), $\{1, 3\}$ (010120102012), and $\{4, 10\}$ (012010201021), each yielding a standard-basis $\mathbb{R}^3$ assignment $i \mapsto [e_{c(i)+1}]$. For the minimum Δ_3 of $\sum |\langle v_i, v_j \rangle|^2$ over the 30 edges $\{i, j\}$ of $\Gamma(G_{32})$ and all unit vectors $v_1, \ldots, v_{15} \in \mathbb{C}^3$, the robustness bounds $1/14400 < \Delta_3 < 0.927496$ are proved in Sec. S3 of the supplementary material. □

In his proof that state-independent contextuality in $\mathbb{C}^3$ requires 13 rays, Cabello[8] found Γ_0 to be the only square-free graph with fewer than 13 vertices and chromatic number above 3, and excluded it by showing, in real coordinates, that ten of its vertices (in our labels, all but 2 and 8) cannot be distinct rays. Part (1) proves this over $\mathbb{C}^3$: the rays $2, 8, 13, 14$ of H_8 are then fixed as orthogonal completions, and the ten rays satisfy $[v_1] = [v_4] = [v_9]$ or $[v_3] = [v_7] = [v_{12}]$. Part (3) shows more: Γ_0 admits no orthogonal ray assignment in $\mathbb{C}^3$ at all, even with repeated rays.

The complete normalized coloring lists for H_8 and $H_{17,10}$ are in the supplement.

Corollary 12 (Blanuša and Isaacs dot-product snarks)**.** *If a cubic graph G admits no orthogonal vector edge coloring in $\mathbb{C}^3$, neither does any Isaacs dot product $G \cdot P$[29], obtained by deleting two nonadjacent edges of G and two adjacent vertices u, v of the Petersen graph P, and joining the two dangling edges of u to the ends of one deleted edge and those of v to the ends of the other. In particular, both 18-vertex Blanuša snarks $B_1, B_2 = P \cdot P$[30] and all iterated dot products $X = P \cdot P \cdots P$ satisfy $\chi(H(X)) = 4 > 3 = \omega(L(X))$ and admit no orthogonal ray assignment in $\mathbb{C}^3$.*

Proof. In $G \cdot P$, the embedded copy of $P \setminus \{u, v\}$ has dual hypergraph H_8, so Theorem 11(1) forces $[v_2] = [v_8]$ and $[v_{13}] = [v_{14}]$ on the two pairs of cut edges, inducing a valid orthogonal ray assignment on G. □

Silva and Ibiapina[23] (Question 6) asked whether a cubic graph with chromatic index 4 can have an orthogonal vector edge coloring in $\mathbb{R}^3$. For the Petersen graph, the Blanuša snarks, and these dot products the answer is no, even in $\mathbb{C}^3$.

C. Completed Yu–Oh H^+_{YO}, base-coloring partition, and context criticality

Completing each of the 12 uncovered orthogonal pairs of the 13-ray Yu–Oh set[10] $(r_1, \ldots, r_{13} \in \mathbb{R}^3$, consisting of $z_1 = r_2, z_2 = r_8, z_3 = r_5$, $y_1^{\pm} = r_{1,3}, y_2^{\pm} = r_{7,9}, y_3^{\pm} = r_{4,6}$, and diagonals $h_0 = r_{12}, h_1 = r_{11}, h_2 = r_{10}, h_3 = r_{13})$ by cross-product rays $r_{14}, \ldots, r_{25}$ (Table S2) yields the 25-ray, 16-context linear hypergraph H^+_{YO} of Berge girth five[7,31]:

$$\begin{aligned} E^+_{\mathrm{YO}} &= \{123, 789, 456, 285\} \cup \bigcup_{h=10}^{13} \mathcal{E}_h, \\ \mathcal{E}_h &= \{\{x, h, w_{xh}\} : x \in N(h)\}, \end{aligned} \tag{15}$$

All 25 vectors, 16 contexts, and restriction-membership flags are tabulated in the supplement. Here the neighborhoods are $N(10) = \{1, 9, 4\}$, $N(11) = \{3, 7, 4\}$, $N(12) = \{3, 9, 6\}$ and $N(13) = \{1, 7, 6\}$, and each w_{xh} is the unique completing ray. It has a real FOR in $\mathbb{R}^3$ (48 orthogonal pairs) and 24 unital separating states (each of the 12 binary assignments on the four base contexts leaves uniquely one diagonal h_j with all three neighbors zero, allowing

$s(h_j) \in \{0,1\}$ and hence $12 + 4 \times 3 = 24$ states). Completed Yu–Oh is therefore a faithfully realized real configuration in which noncontextual truth assignments survive, enough of them to separate every pair of rays, while by the following theorem no noncontextual assignment of three eigenvalues to its outcomes exists.

Theorem 13 (Base-coloring partition and context criticality of H_{YO}^{+})**.** *The completed Yu–Oh hypergraph H_{YO}^{+} satisfies $\omega(\Gamma(H_{\mathrm{YO}}^{+})) = 3$, $\chi_f(\Gamma(H_{\mathrm{YO}}^{+})) = 35/11 > 3 = \chi_f(\Gamma(G_{32}))$, and $\chi(H_{\mathrm{YO}}^{+}) = 4$, and is context-critical for $\chi > 3$:*

1. *Fixing $(c(2), c(8), c(5)) = (0, 1, 2)$, the four base contexts $\{1,2,3\}$, $\{7,8,9\}$, $\{4,5,6\}$, $\{2,8,5\}$ admit eight strong 3-colorings indexed by $(b_1, b_2, b_3) \in \{0,1\}^3$:*

$$\begin{aligned} (c(1), c(3)) &= (2 - b_1, 1 + b_1), \\ (c(7), c(9)) &= (2 - 2b_2, 2b_2), \\ (c(4), c(6)) &= (1 - b_3, b_3). \end{aligned} \tag{16}$$

   Each antipodal pair of (b_1, b_2, b_3) makes the neighborhood $N(h_j)$ of uniquely one *diagonal ray $h_j \in \{10, 11, 12, 13\}$ rainbow ($\{0,1,2\}$) while the other three neighborhoods are 2-colored:*

Antipodal binary patterns	*Blocked diagonal*
$101, 010$	12
$110, 001$	11
$000, 111$	10
$011, 100$	13

(17)

2. *Consequently, $\chi(H_{\mathrm{YO}}^{+}) = 4$, whereas deleting any single context $e \in E_{\mathrm{YO}}^{+}$ yields $\chi(H_{\mathrm{YO}}^{+} \setminus \{e\}) = 3$.*

Proof. Direct check of Eqs. (16)–(17) shows that every base 3-coloring blocks one h_j, so $\chi(H_{\mathrm{YO}}^{+}) \geq 4$, matched by the four-coloring

$$(c(1), \ldots, c(25)) = (2,0,1,1,2,0,2,1,0,3,0,2,1,2,1,1,1,0,0,0,0,2,1,2,0),$$

in which only $h_2 = 10$ receives the fourth color. Deleting a sector context $\{x, h_j, w\}$ frees h_j in its antipodal base colorings by setting $c(h_j) = c(x)$; deleting $\{2,8,5\}$ allows $c(2) = c(8) = c(5) = 0$; deleting $\{1,2,3\}$ allows $(c(1), c(2), c(3)) = (0,0,1)$, $(c(7), c(8), c(9)) = (2,1,0)$, $(c(4), c(5), c(6)) = (1,2,0)$, and $(c(10), c(11), c(12), c(13)) = (2,0,2,1)$; the cases $\{7,8,9\}$ and $\{4,5,6\}$ follow by the threefold cyclic coordinate symmetry of the Yu–Oh set. For χ_f, weight $1/2$ on the 6 states of G_{32} gives $\chi_f(G_{32}) = 3$, while Proposition 7 preserves the 13-ray value $\chi_f = 35/11$[15]. □

IV. EXACT-PALETTE CHROMATIC SEPARABILITY AND LOGICAL GADGETS IN DIMENSIONS THREE AND FOUR

From now on we assume that noncontextual assignments of d eigenvalues exist ($\chi(H) = d$) and ask what they imply for the relations between distinct outcomes. Such assignments may force outcomes to share or to avoid an eigenvalue although ordinary truth assignments permit otherwise; these forced relations are their logical content.

A. Kochen–Specker antecedents and palette-invariant implications

Three historical Kochen–Specker constructions clarify the hierarchy of logical gadgets:

1. *1965 abstract TIFS precursor:* Identifying repeated labels v_1 and v_2 in Kochen and Specker's 1965 Fig. 1[32] (p. 182) yields an 11-vertex, 7-context hypergraph $\{134, 325, 2\,9\,11, 11\,1\,10, 479, 5\,8\,10, 678\}$ implementing $s(1) = 1 \implies s(2) = 0$. It has only four states ($\{1, 5, 6, 9\}$, $\{2, 4, 6, 10\}$, $\{3, 6, 9, 10\}$, $\{4, 5, 6, 11\}$), all with $s(6) = 1$ and $s(7) = s(8) = 0$, so $\chi > 3$; moreover, 1 and 9 are both orthogonal to 4 and 11, so the distinct rays 4 and 11 of an injective realization in $\mathbb{C}^3$ would force $[v_1] = [v_9]$; no injective realization exists (Sec. S3 of the supplementary material).

2. *1967 directed TITS* Γ_1*:* The 10-vertex graph Γ_1 of Lemma 1 in Ref.[4] (pp. 68, 70) implements the one-way implication $s(a_0) = 1 \implies s(a_9) = 1$. Completing its six uncompleted edges by fresh rays $23, 19, 18, 15, 14, 5$ yields the 16-ray, 9-context configuration $I_{16,9}$ in Fig. 2(d) (explicit label map in Sec. S3 of the supplementary material).

3. *1967 state-inseparable* Γ_3*:* In Ref.[4] (p. 70), Γ_3 forces $s(a) = s(b)$ in *every* two-valued state (a, b are nonseparated by $\mathcal{S}(H)$). By contrast, our chromatically inseparable examples below have $\mathcal{S}(H)$ separating all vertices, with $c(u) = c(v)$ forced only on $\mathcal{S}_{\text{ext}}(H)$.

For distinct nonadjacent terminals $u, v \in V$, an ordinary TIFS[33] obeys $s(u) = 1 \implies s(v) = 0$ (equivalently $s(u) + s(v) \leq 1$, which is always bidirectional), and a directed ordinary TITS obeys $s(u) = 1 \implies s(v) = 1$. When $\mathcal{C}_d(H) \neq \varnothing$, an *unanchored chromatic TIFS* obeys $c(u) = 0 \implies c(v) \in \{1, \ldots, d-1\}$, and an *unanchored chromatic TITS* obeys $c(u) = 0 \implies c(v) = 0$.

Proposition 14 (Palette symmetry and purely chromatic gadgets)**.** *For chromatically complete H and nonadjacent $u, v \in V$, the unanchored chromatic TIFS is equivalent to $c(u) \neq c(v)$ for all $c \in \mathcal{C}_d(H)$ (an* implicit edge[11] *(Def. 3.1)), and the unanchored chromatic TITS is equivalent to $u \sim_c v$ (an* implicit identity*). Both chromatic relations are bidirectional. Ordinary TIFS and directed ordinary TITS entail their chromatic counterparts, and with only $c(u) = 0$ fixed, the allowed output set $\{c(v) : c \in \mathcal{C}_d(H),\ c(u) = 0\}$ is one of $\{0\}$, $\{1, \ldots, d-1\}$, or $\{0, \ldots, d-1\}$.*

Proof. Global palette permutations act transitively on $\{0, \ldots, d-1\}$ and, stabilizing 0, transitively on $\{1, \ldots, d-1\}$. Since every color class $s_a(x) = \mathbf{1}_{\{c(x)=a\}}$ is in $\mathcal{S}_{\text{ext}}(H) \subseteq \mathcal{S}(H)$, any ordinary TIFS or one-way TITS forces $c(u) \neq c(v)$ or $c(u) = c(v)$, respectively. □

We call a chromatic TIFS *purely chromatic* if some $s \in \mathcal{S}(H) \setminus \mathcal{S}_{\text{ext}}(H)$ satisfies $s(u) = s(v) = 1$, and a chromatic identity $u \sim_c v$ *two-way purely chromatic* if ordinary states in $\mathcal{S}(H) \setminus \mathcal{S}_{\text{ext}}(H)$ realize both $(s(u), s(v)) = (1, 0)$ and $(0, 1)$, so neither directed ordinary TITS holds. Since a state with $s(u) = s(v) = 1$ cannot be a color class when $c(u) \neq c(v)$ for all c, and a state with $s(u) \neq s(v)$ cannot be one when $u \sim_c v$, the qualifier $\mathcal{S}(H) \setminus \mathcal{S}_{\text{ext}}(H)$ is automatic: a chromatic TIFS is purely chromatic exactly when the ordinary TIFS fails, and a chromatic identity is two-way purely chromatic exactly when both directed ordinary TITS fail.

B. Faithful Yu–Oh restrictions $H_{17,10}$ and $H_{21,13}$ and exhaustive inventory search

The base-coloring partition of H_{YO}^{+} in Theorem 13 directly produces chromatically inseparable faithful restrictions when diagonal sectors are omitted. Retain the four base contexts $\{1, 2, 3\}$, $\{7, 8, 9\}$, $\{4, 5, 6\}$, $\{2, 8, 5\}$ of Eq. (15). Keeping the two diagonal sectors at 10 and 11 defines $H_{17,10}$ on 17 rays and 10 contexts [Fig. 1(a)]; keeping the three sectors at $10, 11, 13$ defines $H_{21,13}$ on 21 rays and 13 contexts [Fig. 1(b)]. Deleting $\{4, 5, 6\}$, $\{4, 11, 24\}$, $\{4, 10, 18\}$ from $H_{17,10}$ recovers the 13-ray, 7-context completed Specker bug $B_{13,7}$ [Fig. 2(b)].

Theorem 15 (Chromatic inseparability in $H_{17,10}$ and $H_{21,13}$)**.** *Both $H_{17,10}$ and $H_{21,13}$ are linear of Berge girth five and inherit injective edge-faithful realizations in $\mathbb{R}^3$ from H_{YO}^{+}:*

(a) $H_{17,10}$ (17 rays, 10 contexts, 2 inseparable pairs)

(b) $H_{21,13}$ (21 rays, 13 contexts, 21 inseparable pairs)

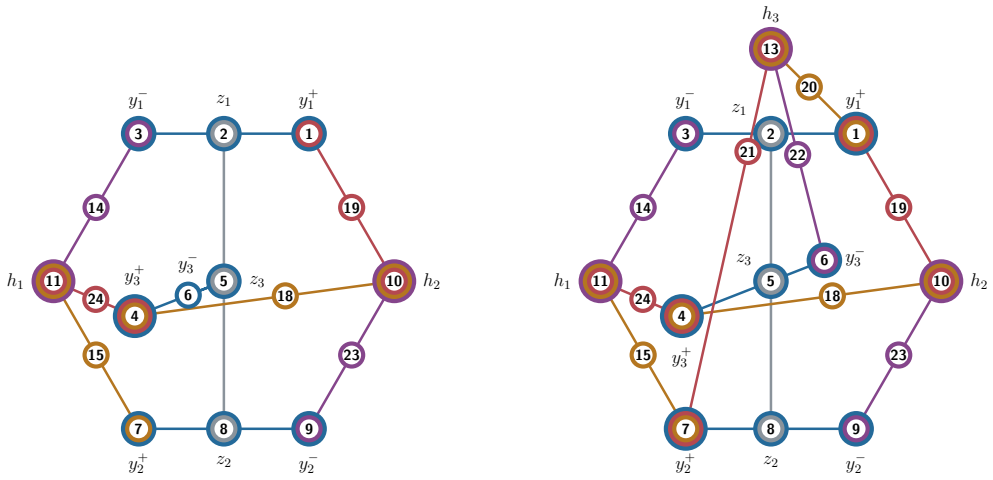


FIG. 1. Faithful three-dimensional Yu–Oh restrictions in $\mathbb{R}^3$, drawn in the common vertex layout of completed Yu–Oh. (a) $H_{17,10}$, omitting diagonal sectors $h_0 = 12$ and $h_3 = 13$, with four normalized strong 3-colorings and chromatically inseparable pairs $10 \sim_c 24$ and $11 \sim_c 18$; (b) $H_{21,13}$, omitting only $h_0 = 12$, with two normalized strong 3-colorings and 21 chromatically inseparable pairs across nine chromatic signature classes.

1. *$H_{17,10}$ has* 18 *unital separating states,* 4 *normalized strong* 3*-colorings (Table S41), and* 10 *extendable states. Its only distinct chromatically inseparable pairs are* $(10, 24)$ *and* $(11, 18)$*. The ordinary implications* $10 \Rightarrow 24$ *and* $11 \Rightarrow 18$ *already hold, but their converses fail; these identities are not two-way purely chromatic. Relative to the anchor* $(c(2), c(8), c(5)) = (0, 1, 2)$*, its four colorings restrict on* $(9, 10, 11)$ *to* $(0, 2, 1)$*,* $(0, 2, 0)$*,* $(2, 1, 2)$*,* $(2, 0, 2)$*, yielding the anchored implications*

$$\begin{aligned} c(9) = 0 &\implies c(11) \in \{0, 1\}, \\ c(10) = 0 &\implies c(11) = 2. \end{aligned} \tag{18}$$

2. *$H_{21,13}$ has* 21 *unital separating states,* 2 *normalized strong* 3*-colorings, and* 6 *extendable states. Its nine chromatic signature classes* $(c_1(v), c_2(v)) \in \{0, 1, 2\}^2$ *are listed in Table S44, giving* $3\binom{4}{2} + 3\binom{2}{2} = 21$ *chromatically inseparable pairs. Exactly* 15 *are two-way purely chromatic identities (Table S70). The remaining six pairs obey the directed ordinary implications*

$$\begin{aligned} &11 \Rightarrow 18, \quad 11 \Rightarrow 21, \quad 13 \Rightarrow 15, \\ &13 \Rightarrow 19, \quad 10 \Rightarrow 20, \quad 10 \Rightarrow 24, \end{aligned}$$

each with a false converse. This classification exhausts all 21 *pairs; no distinct vertices are inseparable under all ordinary states.*

Proof. By Theorem 13(1), among the 8 base colorings in Eq. (16), sector 11 excludes $(1, 1, 0)$, $(0, 0, 1)$ and sector 10 excludes $(0, 0, 0)$, $(1, 1, 1)$. The 4 surviving (b_1, b_2, b_3) extend uniquely to $H_{17,10}$ (Table S41): in each, $N(10) = \{1, 9, 4\}$ and $N(11) = \{3, 7, 4\}$ are 2-colored with $c(10) \neq c(4)$ and $c(11) \neq c(4)$, and $c(10) \neq c(11)$ by the restrictions to $(9, 10, 11)$ listed in (1), forcing $c(24) = c(10)$ in $\{4, 11, 24\}$ and $c(18) = c(11)$ in $\{4, 10, 18\}$. Adding sector 13 in $H_{21,13}$ additionally excludes $(0, 1, 1)$, $(1, 0, 0)$, leaving only $(1, 0, 1)$ and $(0, 1, 0)$, whose column signatures are listed in Table S44. Each retained diagonal adds 3 states to the 12 base states, giving $12 + 2 \times 3 = 18$ and $12 + 3 \times 3 = 21$ states (Sec. S2 of the supplementary material). Testing both directed truth patterns on these 21 states gives the exhaustive 15-versus-6 classification; Table S70 of the supplementary material supplies witnesses for all 15 two-way pairs, together with a reproducible enumeration procedure. □

The full chromatic signatures and both-direction truth witnesses are in the supplement. The witnesses are generated from the complete state families and checked against every context.

Physically, every noncontextual eigenvalue assignment on $H_{17,10}$ gives the nonorthogonal rays 10 and 24, and likewise 11 and 18, the same eigenvalue, although ordinary truth assignments distinguish them. In $H_{21,13}$, fifteen of the twenty-one forced identities are not even shadowed by a one-way truth implication: they exist only at the level of noncontextual eigenvalues.

Proposition 16 (Complete inventory classification)**.** *Among all* $2^{16} - 1 = 65{,}535$ *nonempty context subsets* $E' \subseteq E(H^+_{\mathrm{YO}})$ *with inherited coordinates from Tables S1–S3, exactly* 539 *are connected and edge-faithful, and among these* precisely ten *are* 3*-colorable with a nontrivial chromatic inseparability class: the* $\binom{4}{2} = 6$ *two-sector restrictions (*17 *rays,* 10 *contexts,* 18 *states,* 4 *normalized colorings,* 2 *inseparable pairs) and the* $\binom{4}{3} = 4$ *three-sector restrictions (*21 *rays,* 13 *contexts,* 21 *states,* 2 *normalized colorings,* 21 *inseparable pairs). Thus* 17 *rays and* 10 *contexts are minimal for chromatic inseparability within the faithful restrictions of* H^+_{YO} *with inherited coordinates.*

Proof. Since H^+_{YO} is linear and edge-faithful, a subset E' is edge-faithful if and only if no omitted context contains two vertices of $V' = \bigcup E'$. Exact enumeration by two independent coloring algorithms over all 539 connected faithful subsets certifies the ten solutions listed in Table S71. Disconnected restrictions separate distinct components by independent palette permutations. □

The ten retained-sector cases and binary search masks are enumerated in the supplement. The restriction to inherited coordinates matters: $I_{16,9}$ of Sec. IV C is $H_{17,10}$ without the context $\{4, 5, 6\}$; it has 16 rays and 9 contexts and keeps both identities, but it is edge-faithful only with refitted coordinates.

C. Faithful three-dimensional TIFS, TITS, and purely chromatic gadgets

Figure 2 and Table II synthesize our faithful real 3-dimensional gadgets (coordinates and state/coloring tables in Secs. S1–S2 of the supplementary material):

- *Pentagon* P_5 *[Fig. 2(a)]:* Contexts $\{1, 6, 2\}$, $\{2, 7, 3\}$, $\{3, 8, 4\}$, $\{4, 9, 5\}$, $\{5, 10, 1\}$ have 5 normalized 3-colorings, 11 states, and 10 extendable states, separating all vertex pairs (showing that chromatic separation does not require $\mathcal{S}_{\mathrm{ext}} = \mathcal{S}$). Its degree-one vertices obey the purely chromatic ternary exclusion $\neg(c(6) = c(7) = c(9))$: assigning color 0 to $6, 7, 9$ forces $1, 2, 3, 4, 5$ into $\{1, 2\}$ with $c(1) = c(3)$ and $c(4) = c(5)$, contradicting $\{4, 9, 5\}$, whereas the nonextendable state $\{6, 7, 8, 9, 10\}$ (whose complement is the 5-cycle 12345) assigns 1 to all three terminals.

- *Specker bug* $B_{13,7}$ *and completed* Γ_1 $I_{16,9}$ *[Fig. 2(b),(d)]:* $B_{13,7}$ is an ordinary and chromatic TIFS ($c(10) \neq c(11)$, overlap 1/9). Adjoining $\{4, 11, 24\}$ and $\{4, 10, 18\}$ gives $I_{16,9}$ (deleting $\{4, 5, 6\}$ from $H_{17,10}$ with refitted faithful coordinates), where $c(10) \neq c(11)$ forces $c(4)$ to be the third color, so $c(24) = c(10)$ and $c(18) = c(11)$ via the one-way ordinary TITS $s(10) = 1 \implies s(24) = 1$ (overlap 25/27).

Proposition 17 (Purely chromatic binary TIFS $D_{16,9}$ and two-way identity $T_{19,11}$)**.** *Let* $D_{16,9} = H_{17,10} \setminus \{\{1, 10, 19\}\}$ *(with orphan* 19 *removed) and* $T_{19,11} = D_{16,9} \cup \{\{14, 26, 27\}, \{18, 26, 28\}\}$*, realized faithfully in* $\mathbb{R}^3$ *[Fig. 2(c),(e)]:*

1. *Every* $c \in \mathcal{C}_3(D_{16,9})$ *satisfies* $c(14) \neq c(18)$ *(*7 *normalized colorings,* 14 *extendable states), whereas the nonextendable state* $\{1, 5, 7, 14, 18, 23, 24\} \in \mathcal{S}(D_{16,9})$ *assigns* $s(14) = s(18) = 1$ *(squared terminal overlap* 16/2025*).*

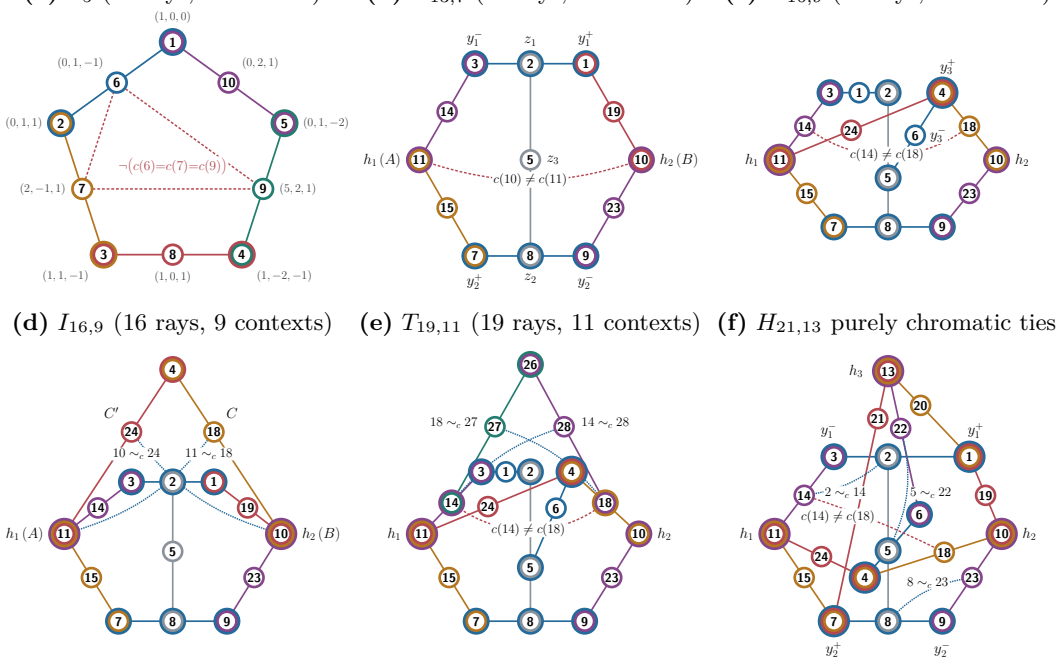


FIG. 2. Faithful real three-dimensional logical gadgets in $\mathbb{R}^3$. (a) Pentagon P_5 with purely chromatic ternary exclusion $\neg(c(6) = c(7) = c(9))$; (b) completed Specker bug $B_{13,7}$ (ordinary TIFS $c(10) \neq c(11)$); (c) purely chromatic binary TIFS $D_{16,9}$ ($c(14) \neq c(18)$); (d) completed historical Γ_1 TITS $I_{16,9}$ ($10 \sim_c 24, 11 \sim_c 18$); (e) two-way purely chromatic identity $T_{19,11}$ ($14 \sim_c 28, 18 \sim_c 27$); (f) $H_{21,13}$ showing three of its fifteen two-way purely chromatic identities: $2 \sim_c 14, 5 \sim_c 22, 8 \sim_c 23$. Red triangles forbid monochromatic triples; red dashed arrows denote $c(u) \neq c(v)$; blue dotted arrows denote $u \sim_c v$.

2. *Every* $c \in \mathcal{C}_3(T_{19,11})$ *satisfies* $14 \sim_c 28$ *and* $18 \sim_c 27$ *(7 normalized colorings, 25 states, 14 extendable), while for both pairs, nonextendable states* $\{1, 5, 7, 14, 18, 23, 24\}$ *and* $\{1, 6, 8, 10, 11, 27, 28\}$ *realize* $(1, 0)$ *and* $(0, 1)$*, violating both directed ordinary TITS implications (squared overlap* 2009/2025*; Table S70).*

Proof. (1) If $c(14) = c(18) = 0$, set $c(3) = 1, c(11) = 2$ by $\{3, 11, 14\}$. Then $\{4, 11, 24\}$ and $\{4, 10, 18\}$ force $c(4) = 1$ and $c(10) = 2$. In $\{7, 8, 9\}$, $7 \perp 11$ and $9 \perp 10$ avoid color 2, forcing $c(8) = 2$. Then $\{1, 2, 3\}$ and $\{4, 5, 6\}$ make 2 and 5 avoid color 1, so $\{2, 8, 5\}$ forces $c(2) = c(5) = 0$, a contradiction. (2) In $T_{19,11}$, $c(14) \neq c(18)$ forces their common neighbor 26 ($r_{26} \parallel r_{14} \times r_{18}$) to take the third color, whence $c(27) = c(18)$ and $c(28) = c(14)$. □

D. Apex extensions and faithful four-dimensional linear gadgets

Proposition 18 (Apex extension)**.** *Embedding any* 3*-uniform hypergraph* H *realized in* $\mathbb{F}^3$ *into* $\mathbb{F}^d$ *(*$d \geq 3$*) and appending* $d - 3$ *mutually orthogonal coordinate apices to every context preserves injectivity, edge-faithfulness,* $\chi(H^{(d)}) = \chi(H) + d - 3$*, normalized coloring count, and old chromatic inseparability classes (with apices as singletons), while adding* $d - 3$ *apex-only states (all extendable when* $\chi(H) = 3$*) to the zero-extended states of* H*.*

Proof. In every coloring of $H^{(d)}$, the $d - 3$ apices occupy $d - 3$ distinct colors across all contexts, leaving a 3-coloring of H; similarly, any state either sets one apex to 1 (forcing all old vertices to 0) or sets all apices to 0 and restricts to $\mathcal{S}(H)$. □

TABLE II. Faithful geometric gadgets. $N_c = |\mathcal{C}_d|/d!$ fixes one context palette. Terminal relations describe colors unless explicitly identified as ordinary truth implications.

Gadget	d	$\lvert V\rvert/\lvert E\rvert$	N_c	$\lvert\mathcal{S}\rvert$	$\lvert\mathcal{S}_{\text{ext}}\rvert$	Relation and mechanism
P_5	3	10/5	5	11	10	Ternary exclusion on $6, 7, 9$; purely chromatic.
$B_{13,7}$	3	13/7	6	14	12	$c(10) \neq c(11)$; ordinary TIFS.
$D_{16,9}$	3	16/9	7	21	14	$c(14) \neq c(18)$; purely chromatic TIFS.
$I_{16,9}$	3	16/9	6	22	12	$10 \sim_c 24$, $11 \sim_c 18$; directed ordinary TITS.
$T_{19,11}$	3	19/11	7	25	14	$14 \sim_c 28$, $18 \sim_c 27$; both truth directions fail.
$H_{21,13}$	3	21/13	2	21	6	21 identical pairs: 15 two-way purely chromatic; six directed ordinary TITS.
$H_{17,10}$	3	17/10	4	18	10	Two color-identical pairs from directed ordinary TITS.
$W^{(4)}_{11,5}$	4	11/5	5	12	11	Apex ternary exclusion.
$W^{(4)}_{16,6}$	4	16/6	80	72	56	Linear ternary exclusion on $6, 8, 11$.
$B^{(4)}_{14,7}$	4	14/7	6	15	13	Apex ordinary TIFS.
$L^{(4)}_{18,7}$	4	18/7	144	68	68	Linear ordinary TIFS $c(11) \neq c(15)$.
$D^{(4)}_{17,9}$	4	17/9	7	22	15	Apex purely chromatic TIFS.
$L^{(4)}_{22,9}$	4	22/9	216	158	92	Linear identity $11 \sim_c 20$ from directed ordinary TITS.

Applying Proposition 18 at $d = 4$ to P_5, $B_{13,7}$, and $D_{16,9}$ yields the non-linear ($g = 2$) four-dimensional gadgets $W^{(4)}_{11,5}$, $B^{(4)}_{14,7}$, and $D^{(4)}_{17,9}$ in Fig. 3(a),(c),(e). To achieve *linearity* ($|e \cap e'| \leq 1$) in $\mathbb{R}^4$, we construct three faithful linear gadgets ($g = 3$) in Fig. 3(b),(d),(f) (Table II):

1. *Linear purely chromatic ternary gadget* $W^{(4)}_{16,6}$ *[Fig. 3(b)]:* Contexts $\{2, 0, 6, 1\}$, $\{2, 7, 13, 3\}$, $\{3, 0, 8, 4\}$, $\{4, 9, 14, 5\}$, $\{5, 10, 15, 1\}$, $\{0, 11, 12, 5\}$ satisfy $\neg(c(6) = c(8) = c(11))$ (if $c(6) = c(8) = c(11) = 0$ and $c(0) = 3$, then $1, 2, 3, 4, 5$ must 2-color a 5-cycle), while $\{6, 7, 8, 9, 10, 11\} \in \mathcal{S}(W^{(4)}_{16,6})$ sets all three terminals to 1.

2. *Linear ordinary TIFS* $L^{(4)}_{18,7}$ *[Fig. 3(d)]:* Contexts $\{1, 2, 4, 3\}$, $\{5, 8, 7, 6\}$, $\{4, 9, 10, 8\}$, $\{11, 1, 12, 5\}$, $\{11, 14, 13, 2\}$, $\{15, 3, 16, 7\}$, $\{15, 17, 18, 6\}$ enforce $s(11) = 1 \implies s(15) = 0$ and $c(11) \neq c(15)$ (if $c(11) = c(15) = a$, then $1, 2, 3$ and $5, 6, 7$ avoid a, forcing $c(4) = c(8) = a$ in $\{4, 9, 10, 8\}$). Using $r_{13} = (1, 0, 1, 1), r_{14} = (2, 0, -1, -1)$ avoids an unlisted orthogonality $13 \perp 15$; all 68 states are extendable (terminal overlap 1/42).

3. *Linear directed TITS and color identity* $L^{(4)}_{22,9}$ *[Fig. 3(f)]:* Adjoining $\{11, 21, 22, 19\}$ and $\{13, 20, 15, 19\}$ to $L^{(4)}_{18,7}$ (with $r_{13} = (1, 0, 4, 4), r_{14} = (8, 0, -1, -1)$ so $r_{13} \perp r_{15}$) makes $\{11, 13, 15, 19\}$ pairwise differently colored, forcing $11 \sim_c 20$ and $s(11) = 1 \implies s(20) = 1$ (while $\{1, 6, 9, 14, 16, 20, 21\}$ has $s(11) = 0, s(20) = 1$; terminal overlap 41/42).

V. THE 37-RAY COMPLEX HESSE HYPERGRAPH AND RATIONAL FOUR-DIMENSIONAL COMPLETIONS

A. The 37-ray complex Hesse hypergraph H_{37} and its 76-event orthomodular lattice

With $\zeta = \exp(2\pi i/3)$ ($\zeta^2 + \zeta + 1 = 0$) and $a, k \in \mathbb{Z}_3$, define 37 rays in $\mathbb{C}^3$ (core rays z, M_{ak}, P_k, Q_k and unique cross-product completion rays A_k, B_{ak}, C_{ak}, numbered $1, \ldots, 37$

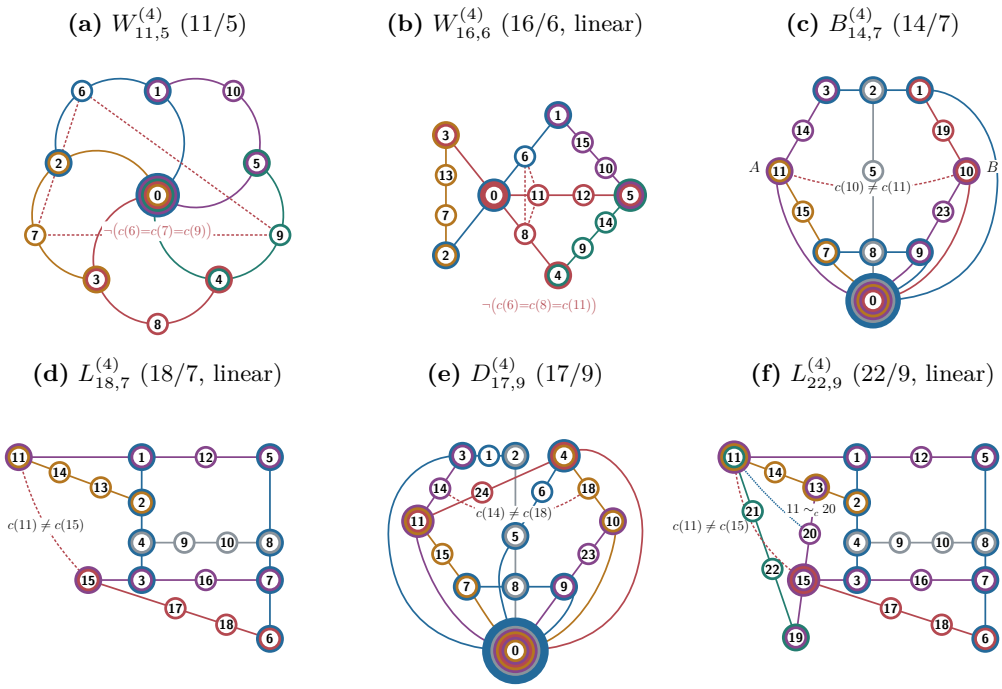


FIG. 3. Faithful real four-dimensional gadgets in $\mathbb{R}^4$. Panels (a),(c),(e) are common-apex lifts ($g(H) = 2$); panels (b),(d),(f) are linear ($g(H) = 3$). (a),(b) Purely chromatic ternary exclusions; (c),(d) ordinary TIFS; (e) purely chromatic binary TIFS $c(14) \neq c(18)$; (f) linear color identity $11 \sim_c 20$ from directed ordinary TITS $s(11) = 1 \implies s(20) = 1$.

via $z \leftrightarrow 1, M_{ak} \leftrightarrow 2{+}3a{+}k, P_k \leftrightarrow 11{+}k, Q_k \leftrightarrow 14{+}k, A_k \leftrightarrow 17{+}k, B_{ak} \leftrightarrow 20{+}3a{+}k, C_{ak} \leftrightarrow 29 + 3a + k$):

$$\begin{aligned} z &= (1,0,0), & M_{ak} &= (1, \zeta^{a+k}, \zeta^{a+2k}), \\ P_k &= (0,1,-\zeta^k), & Q_k &= (-\zeta^k, 0, 1), \\ A_k &= (0,1,\zeta^k), & B_{ak} &= (-2, \zeta^{a+k}, \zeta^{a+2k}), \\ C_{ak} &= (1, -2\zeta^{2a+k}, \zeta^{2k}), \end{aligned} \tag{19}$$

and the 24 complete orthogonal triad contexts (Fig. 4):

$$\begin{aligned} \mathcal{M}_a &= \{M_{a0}, M_{a1}, M_{a2}\}, & \mathcal{A}_k &= \{z, P_k, A_k\}, \\ \mathcal{B}_{ak} &= \{M_{ak}, P_k, B_{ak}\}, & \mathcal{C}_{ak} &= \{M_{a,a+k}, Q_k, C_{ak}\}. \end{aligned} \tag{20}$$

Theorem 19 (Properties of H_{37})**.** *The hypergraph H_{37} defined by Eqs. (19)–(20) satisfies:*

1. *It is linear with Berge girth five,* 72 *edges, and $\omega(\Gamma(H_{37})) = 3$; its triangles are precisely its* 24 *contexts.*

2. *Equation (19) defines an injective edge-faithful representation in $\mathbb{C}^3$, whereas H_{37} admits* no *injective orthogonal representation in $\mathbb{R}^3$.*

3. *Its chromatic numbers are $\chi(H_{37}) = 4 > 3 = \chi_f(\Gamma(H_{37})) = \omega(\Gamma(H_{37}))$.*

4. *Its* 24 *Boolean blocks paste into a* 76*-event orthomodular lattice $L(H_{37}) = \{0,1\} \cup \{a_i, a_i^{\perp} : 1 \le i \le 37\}$. Its full state space has $|\mathcal{S}(H_{37})| = 153$ states and is order determining on $L(H_{37})$, with exact minimum state-family sizes* 6 *for unital vertex separation,* 7 *for separation of all* 76 *events, and* 18 *for order determination.*

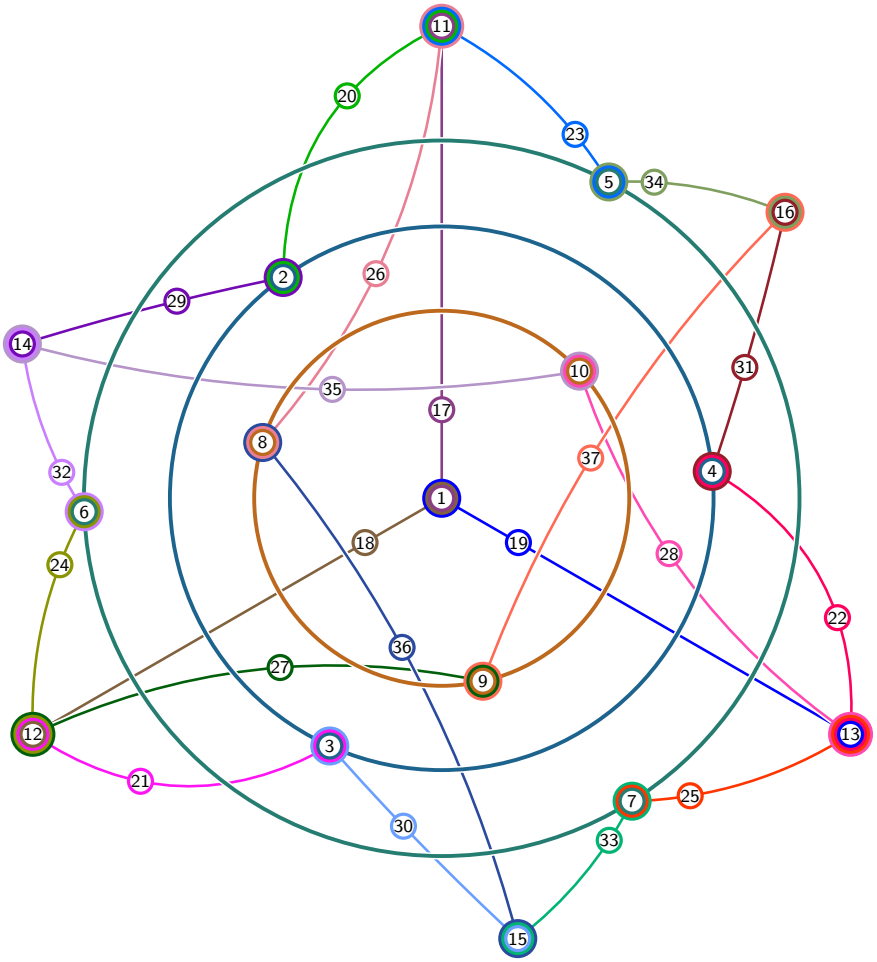


FIG. 4. Complete 3-uniform hypergraph diagram of H_{37} (37 rays, 24 orthogonal triads, 72 edges in $\mathbb{C}^3$). Bold circles denote e_1, e_2, e_3 $(\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2)$; the other 21 smooth traces join core pairs through their unique completion rays $17, \ldots, 37$. Concentric rings record incident context colors.

Proof. (1)–(2) Zero patterns, magnitudes, and ζ-phases distinguish all 37 rays. Algebraic inner products confirm that $\langle v_i, v_j \rangle = 0$ holds strictly for the $24\binom{3}{2} = 72$ context pairs: $\langle M_{ak}, M_{c\ell} \rangle = 1 + \zeta^{c-a}(\zeta^{\ell-k} + \zeta^{2(\ell-k)}) = 0 \iff (c = a, \ell \neq k)$; $P_j \perp M_{ak} \iff j = k$; $Q_j \perp M_{ak} \iff k = a + j$; $\langle M_{c\ell}, B_{ak} \rangle = -2 + \zeta^{a-c}(\zeta^{k-\ell} + \zeta^{2(k-\ell)}) = 0 \iff (c, \ell) = (a, k)$; $\langle M_{c\ell}, C_{ak} \rangle = 1 - 2\zeta^{2a+k-c-\ell} + \zeta^{2k-c-2\ell} = 0 \iff (c = a, \ell = a + k)$; and unequal term magnitudes exclude orthogonalities among A, B, C. Uniqueness of cross-product completions in $\mathbb{C}^3$ excludes Berge 3- and 4-cycles, while $(\mathcal{A}_0, \mathcal{B}_{00}, \mathcal{M}_0, \mathcal{B}_{01}, \mathcal{A}_1)$ is a Berge 5-cycle.

To exclude any injective representation in $\mathbb{R}^3$, suppose unit $z \in \mathbb{R}^3$ and real $p_k, m_{ak} \in \mathbb{R}^3$ realize $\mathcal{A}_k, \mathcal{B}_{ak}, \mathcal{M}_a$. If $m_{ak} \cdot z = 0$, then $\{z, p_k, m_{ak}\}$ is an orthogonal basis of $\mathbb{R}^3$, and for $\ell \neq k$, $p_\ell \perp z$ and $m_{a\ell} \perp m_{ak}$ with $p_\ell \perp m_{a\ell}$ force $[p_\ell] = [m_{ak}]$ or $[m_{a\ell}] = [z]$. Hence $m_{ak} \cdot z \neq 0$. Rescaling so $m_{ak} \cdot z = 1$ and choosing unit $u_k \in z^\perp \cap p_k^\perp$ gives $m_{ak} = z + t_{ak} u_k$ $(t_{ak} \in \mathbb{R} \backslash \{0\})$. Orthogonality of $\mathcal{M}_a$ yields $1 + t_{ak} t_{a\ell}(u_k \cdot u_\ell) = 0$ $(k \neq \ell)$. Setting $q_k = t_{ak}/t_{0k}$ gives $q_0 q_1 = q_0 q_2 = q_1 q_2 = 1$, whose only real solutions are $(q_0, q_1, q_2) = \pm(1, 1, 1)$. Thus at most two distinct bases $\mathcal{M}_a$ exist in $\mathbb{R}^3$, contradicting injectivity on the three bases $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2$. The argument uses only the 13 rays z, P_k, M_{ak} and the orthogonalities among them, so already the 13-vertex subgraph of $\Gamma(H_{37})$ induced by these rays has no injective orthogonal representation in $\mathbb{R}^3$.

(3) In any 3-coloring with $c(z) = 0$, $P_0, P_1, P_2 \in \{1, 2\}$ cannot be monochromatic (as

$\mathcal{M}_a$ uses all three colors and $M_{ak} \sim P_k$). Up to symmetry $(c(P_0), c(P_1), c(P_2)) = (2, 1, 1)$, forcing $c(M_{a0}) = 1$ and $(c(M_{a1}), c(M_{a2})) = (x_a, 2 - x_a)$ with $x_a \in \{0, 2\}$. Since $N(Q_k) = \{M_{0k}, M_{1,k+1}, M_{2,k+2}\}$ avoids a color, $x_1 = 2 - x_2$, $x_0 = 2 - x_1$, $x_2 = 2 - x_0$, impossible on a 3-cycle. Thus $\chi(H_{37}) = 4$ (attained by $c(z) = 0, c(M_{ak}) = k, c(P_k) = c(Q_k) = 3$). The 6-state double cover $\mathcal{R}_6 = \{s_2, s_{30}, s_{73}, s_{99}, s_{109}, s_{139}\}$ covers every vertex twice, giving $\chi_f(\Gamma(H_{37})) = 3$.

(4) Each state is fixed by $t = (t_0, t_1, t_2) \in \mathbb{Z}_3^3$ with $s(M_{ak}) = \mathbf{1}_{k=t_a}$, $s(z) = \varepsilon \in \{0, 1\}$, and independent bits on $U(t) = \mathbb{Z}_3 \setminus \{t_0, t_1, t_2\}$ (for $\varepsilon = 0$) and $W(t) = \mathbb{Z}_3 \setminus \{t_a - a\}$, yielding $(1 + 2^{|U(t)|})2^{|W(t)|}$ states per t:

Class of $t \in \mathbb{Z}_3^3$	$\lvert\#t\rvert$	$(\lvert U(t)\rvert, \lvert W(t)\rvert)$	$(1 + 2^{\lvert U\rvert})2^{\lvert W\rvert}$
(c, c, c)	3	$(2, 0)$	$5 \times 1 = 5$
$(c, c+1, c+2)$	3	$(0, 2)$	$2 \times 4 = 8$
$(c, c+2, c+1)$	3	$(0, 0)$	$2 \times 1 = 2$
Two equal entries	18	$(1, 1)$	$3 \times 2 = 6$

(21)

and hence $|\mathcal{S}(H_{37})| = 3 \cdot 5 + 3 \cdot 8 + 3 \cdot 2 + 18 \cdot 6 = 153$. Since $2^5 - 1 = 31 < 37$ and $2^6 = 64 < 76$, the explicit families Ω_6 and Ω_7 given in the supplement prove the minima 6 and 7. For order determination ($a_i \le a_j^\perp \iff i \sim j$), the 15 nonorthogonal pairs in $D = \{P_0, P_1, P_2, Q_0, Q_1, Q_2\}$ require 15 disjoint joint-one witnesses in $T \subset \mathcal{S}(H_{37})$ ($|T| = 27$), and the 9 pairs $\{z, C_{ak}\}$ require at least 2 states outside T. Exhaustive bit-set verification over all $4{,}096 \times 132 = 540{,}672$ candidate 17-state families shows each meets at most 75 of the 81 minimal witness sets W_{xy}, whereas the 18-state family Ω_{18} in the supplement meets all 81 (Sec. S3 of the supplementary material). □

Theorem 19(2) is not the first finite separation of complex from real three-dimensional orthogonality: Harding and Salinas Schmeis[34] (Thm. 2.6) construct, from mutually unbiased bases, a two-gadget configuration that is realizable in $\mathbb{C}^3$ but not in $\mathbb{R}^3$ (48 rays, 32 bases, 189 context states, and 180 geometric states). H_{37} is smaller, linear of Berge girth five, and edge-faithful in $\mathbb{C}^3$, and its real obstruction already resides in the 13-ray subgraph on z, P_k, M_{ak}.

By Proposition 18, appending $d - 3$ common apices to H_{37} in $\mathbb{C}^d$ ($d \ge 3$) gives faithful d-uniform hypergraphs with $|V| = d + 34$, $|E| = 24$, $\omega = d$, $\chi = d + 1$, and $|\mathcal{S}| = d + 150$.

An order-determining family represents the specified 76-event order, orthocomplements, and Boolean operations within listed contexts. It does not preserve arbitrary meets as set intersections: nonorthogonal atoms have meet zero although some states make both true. It is therefore not a Boolean-sublattice embedding of the full lattice, nor a claim of closure under every Hilbert operation. In particular, the minimum six-state separating family and the six-state uniform double cover solve different problems.

B. Hyperplane obstruction, two-plane completion lemma, and canonical linear H_{65} in $\mathbb{R}^4$

Appending a single apex $v_{26} = (0, 0, 0, 1)^T \in \mathbb{R}^4$ to H_{YO}^+ gives the 26-ray, 16-context non-linear suspension $H_{\mathrm{YO}}^{(4)}$ (Fig. 5) with $\omega = 4, \chi = 5$, and 25 unital separating states (Proposition 18), but $|e \cap e'| = 2$ ($g = 2$). Restricted to the 13 Yu–Oh rays and the apex, its orthogonality graph is the Yu–Oh graph joined to one common vertex, the example that Cabello, Kleinmann, and Budroni[15] (Thm. 1) used to show that $\chi_f > d$ is necessary but not sufficient for state-independent contextuality; H_{65} below is a linear, apex-free example of the same phenomenon.

Proposition 20 (Hyperplane obstruction to linearity in $\mathbb{F}^4$)**.** *In any injective realization in $\mathbb{P}(\mathbb{F}^4)$, if two contexts e, e' each have three rays in a common 3-dimensional hyperplane $W = z^\perp$, both must contain $[z]$; hence if $|e_W \cap e'_W| \ge 1$, then $|e \cap e'| \ge 2$, violating linearity.*

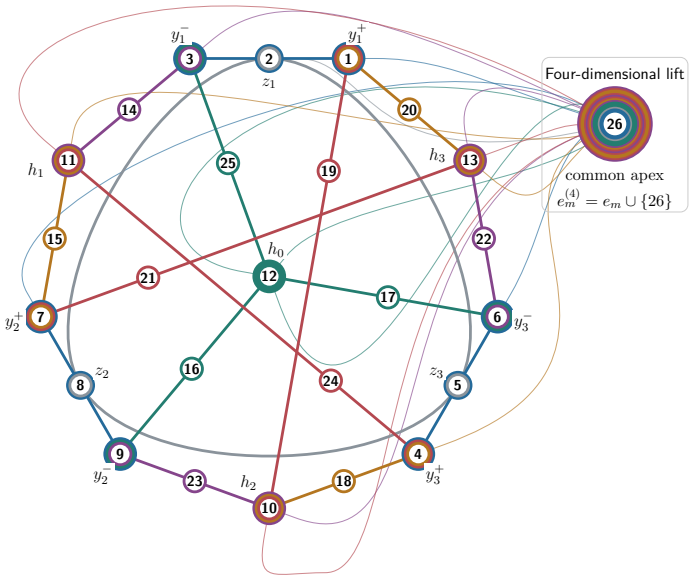


FIG. 5. The completed 25-ray, 16-context Yu–Oh hypergraph H_{YO}^{+} in $\mathbb{R}^3$ ($\omega = 3, \chi = 4$) and its non-linear 26-ray, 16-context apex suspension $H_{\mathrm{YO}}^{(4)}$ in $\mathbb{R}^4$ ($\omega = 4, \chi = 5, g = 2$) obtained by adjoining the common orthogonal apex $v_{26} = (0, 0, 0, 1)^T$ to every context.

Proof. Three mutually orthogonal rays in W span W, forcing the fourth orthogonal ray of both e and e' to equal $[z] = W^{\perp}$. □

Lemma 21 (Simultaneous two-plane completion in $\mathbb{R}^4$)**.** *Let $V_{\mathrm{core}} \subset \mathbb{Q}^4 \setminus \{0\}$ be a finite set of non-collinear rational vectors with 4-element orthogonal base contexts E_{base} ($|C \cap C'| \leq 1$) and let all remaining orthogonal pairs be the bridge pairs $E_{\mathrm{bridge}} = \{e_1, \ldots, e_M\}$ ($e_m = \{u_m, v_m\}$, $L_m = \mathrm{span}(v_{u_m}, v_{v_m})$, $P_m = L_m^{\perp}$) such that every core vertex lies in a base context or is a bridge endpoint.*

1. *The pairs e_m can be simultaneously completed to 4-bases $E_m = \{u_m, v_m, w_{m,1}, w_{m,2}\}$ by fresh integer vectors $w_{m,1}, w_{m,2} \in P_m \cap \mathbb{Z}^4$ forming a linear edge-faithful hypergraph H if and only if*

$$\begin{aligned} v_z &\notin L_m \quad (z \in V_{\mathrm{core}} \setminus e_m), \\ P_\ell &\neq L_m \quad (\ell \neq m). \end{aligned} \tag{22}$$

2. *$\omega(\Gamma(H)) = 4$ and $\chi(H) = \max\{4, \chi(G_{\mathrm{core}})\}$; here G_{core} is automatically K_5-free, since no five nonzero vectors of $\mathbb{R}^4$ are mutually orthogonal.*

3. *If all G_{core}-independent base supports $S_1, \ldots, S_N$ form a unital separating family on the core, and the zero-profiles $Z(e_m) = \{\alpha : S_\alpha \cap e_m = \varnothing\}$ are nonempty and pairwise distinct, the $\leq 2N$ canonical extensions $s_{\alpha,\varepsilon}$ ($\varepsilon \in \{0, 1\}$, setting $(s_{\alpha,\varepsilon}(w_{m,1}), s_{\alpha,\varepsilon}(w_{m,2})) = (1 - \varepsilon, \varepsilon)$ when $\alpha \in Z(e_m)$ and $(0, 0)$ otherwise) are unital and separating on $V(H)$.*

Proof. If $v_z \in L_m$ then $P_m \subset v_z^{\perp}$, and if $P_\ell = L_m$ then $P_\ell \perp P_m$, forcing extra orthogonalities. Conversely, given Eq. (22), an integer orthogonal basis $\{\eta_A, \eta_B\} \subset P_m \cap \mathbb{Z}^4$ defines the infinite family $w_{m,1}(a, b) = a\eta_A + b\eta_B$, $w_{m,2}(a, b) = -\|\eta_B\|^2 b\eta_A + \|\eta_A\|^2 a\eta_B$, avoiding inductively the finitely many 1-dimensional intersections $z^{\perp} \cap P_m$ and $\mathbb{R}z \cap P_m$, where z ranges over the core rays *and* the fresh rays placed at earlier steps, together with $P_m \cap L_r$ ($r > m$). The last condition keeps every fresh ray outside each later L_r, so that $z^{\perp} \cap P_r$ remains 1-dimensional for fresh z as well. Linearity, coloring extension, and distinctness of canonical vertex profiles follow from Proposition 7 and distinctness of $Z(e_m)$. □

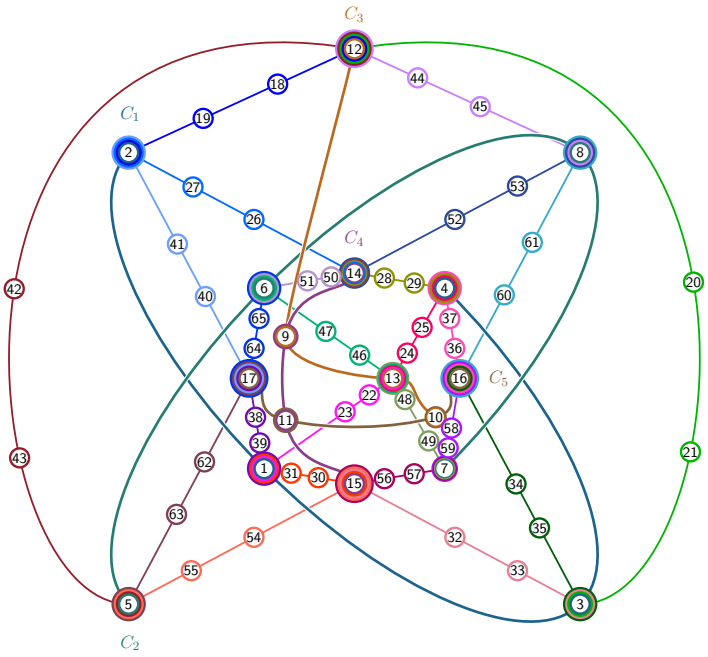


FIG. 6. Canonical linear 4-uniform hypergraph H_{65} in $\mathbb{R}^4$ (65 rays, 29 contexts, 174 edges, $g = 3, \omega = 4, \chi = 5$). Thick colored curves denote the five core contexts $C_1, \ldots, C_5$; the 24 thinner lines denote the completed bridge contexts $E_1, \ldots, E_{24}$ carrying degree-one completion pairs $\{18, 19\}, \ldots, \{64, 65\}$.

To construct a *linear* 4-uniform hypergraph in $\mathbb{R}^4$, we take the 17-ray core $V_{\text{core}} = \{1, \ldots, 17\} \subset \{-1, 0, 1\}^4$, which is orthogonally equivalent to the 17-ray Peres restriction of Haynes *et al.*[19] (pp. 8–9) via the orthogonal matrix

$$T = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & -1 & -1 & -1 \end{pmatrix}, \qquad T^T T = I_4. \tag{23}$$

The core consists of the coordinate basis C_1, an even-parity Hadamard basis C_2, three odd-parity hub rays $9, 10, 11$, and the three binary bridge pairs $B_1 = \{12, 13\}$, $B_2 = \{14, 15\}$, $B_3 = \{16, 17\}$. The remaining base contexts are $C_3 = \{9, 10, 12, 13\}$, $C_4 = \{9, 11, 14, 15\}$ and $C_5 = \{10, 11, 16, 17\}$. Complete coordinates and all compatible supports are in Tables S17–S18 and S81 of the supplementary material. Completing its 24 uncovered bridge edges $e_m \in (C_1 \cup C_2) \times (B_1 \cup B_2 \cup B_3)$ by fresh integer pairs $\{2m+16, 2m+17\} \subset \{18, \ldots, 65\}$ via Lemma 21 yields $H_{65} = (V_{65}, E_{29})$ (Fig. 6; Tables S17–S18 of the supplementary material):

$$\begin{aligned} E_{29} &= \{C_1, \ldots, C_5\} \cup \{E_m : 1 \le m \le 24\}, \\ E_m &= \{u_m, v_m, 2m + 16, 2m + 17\}. \end{aligned} \tag{24}$$

Theorem 22 (Properties of H_{65})**.** *The 65-ray, 29-context hypergraph H_{65} satisfies:*

1. *It is linear ($g(H_{65}) = 3$) and edge-faithful in $\mathbb{R}^4$ (174 edges) with $\omega(\Gamma(H_{65})) = 4$, and nonconformal (13 uncovered maximal triangles: 12 bridge triangles plus $\{9, 10, 11\}$).*

2. *It has $\alpha(G_{\text{core}}) = 4$, $\chi_f(\Gamma(H_{65})) = 17/4 = 4.25 > 4$, and $\chi(H_{65}) = 5$.*

TABLE III. Bridge neighbors in the two core bases. The complete 24 state supports are supplementary.

b	$N(b)\cap C_1$	$N(b)\cap C_2$
12	$\{2,3\}$	$\{5,8\}$
13	$\{1,4\}$	$\{6,7\}$
14	$\{2,4\}$	$\{6,8\}$
15	$\{1,3\}$	$\{5,7\}$
16	$\{3,4\}$	$\{7,8\}$
17	$\{1,2\}$	$\{5,6\}$

3. *Its core has* 24 *compatible state supports* $S_\alpha=\{i_1,i_2,h,b\}$ *(Table S81), each missing* 14 *bridges and yielding* $|\mathcal{S}(H_{65})|=24\times 2^{14}=393{,}216$ *states and* 48 *canonical separating states.*

4. *Every* $s\in\mathcal{S}(H_{65})$ *obeys* $s(9)+s(10)+s(11)=1$*, so adjoining the fourth hub ray* $q=(1,1,1,-1)^T$ *forces* $\widetilde{s}(q)=0$*; moreover,* $A_9=A_{16}\dot\cup A_{17}$ *on all of* $\mathcal{S}(H_{65})$*, forcing the false atom-order implication* $s(16)\le s(9)$ *and collapsing the* 1*-ray* $[v_9]=[v_{16}-v_{17}]$ *with the* 2*-plane* $\mathrm{span}(v_{16},v_{17})$*.*

Proof. (1)–(2) No K_4 in G_{core} contains a bridge edge $\{u_m,v_m\}$, since $u_m\in C_a$ and $v_m\in B_k$ have exactly one common neighbor in G_{core}, namely the second vertex of $N(v_m)\cap C_a$; this verifies $P_\ell\neq L_m$, and support-size / magnitude checks verify $v_z\notin L_m$ in Lemma 21. For any independent $I\subseteq V_{\mathrm{core}}$, $|I\cap C_1|\le 1$ and $|I\cap C_2|\le 1$. If $I\cap\{9,10,11\}\neq\varnothing$, $|I\cap(C_3\cup C_4\cup C_5)|\le 2$ so $|I|\le 4$; if $I\cap\{9,10,11\}=\varnothing$ and I picks $(b_1,b_2,b_3)\in B_1\times B_2\times B_3$, even-parity triples cover C_2 and odd-parity triples cover C_1 (Table III), so again $|I|\le 3{+}1=4$. Thus $\alpha(G_{\mathrm{core}})=4$, $\chi_f\ge 17/4$ (matched by weights $1/8$ on the 24 core states and $1/16$ on the 20 remaining independent 4-sets), and $\chi(H_{65})=5$. (3)–(4) By parity, no state can avoid $\{9,10,11\}$, giving $s(9)+s(10)+s(11)=1$, $3\times 2\times 2\times 2=24$ core supports (Table S81), $24-10=14$ missed bridges per support, and $A_9=A_{16}\dot\cup A_{17}$. □

VI. EVENT ORDER, QUANTUM PROBABILITIES, AND NONEXTENDABLE WEIGHT

A. Extendable state polytopes and nonextendable weight on $H_{17,10}$

For any context hypergraph H, the classical state polytope $\mathcal{P}(H)=\mathrm{conv}\,\mathcal{S}(H)$ and the extendable state polytope $\mathcal{P}_{\mathrm{ext}}(H)=\mathrm{conv}\,\mathcal{S}_{\mathrm{ext}}(H)$ satisfy $\mathcal{P}_{\mathrm{ext}}(H)\subseteq\mathcal{P}(H)$, with $p_u=p_v$ for all $p\in\mathcal{P}_{\mathrm{ext}}(H)$ whenever $u\sim_c v$ by Corollary 4. For $p\in\mathcal{P}(H)$, define its *minimum nonextendable weight*

$$\begin{aligned}\eta_H(p)=\min_{\mu}\quad & \sum_{s\notin\mathcal{S}_{\mathrm{ext}}}\mu_s,\\ \text{subject to}\quad & p=\sum_{s\in\mathcal{S}}\mu_s s,\\ & \mu_s\ge 0,\qquad \sum_s\mu_s=1.\end{aligned}\tag{25}$$

Proposition 23 (Nonextendable weight lower bound)**.** *For any chromatically inseparable pair* $u\sim_c v$ *in* H *and any* $p\in\mathcal{P}(H)$*,* $\eta_H(p)\ge|p_u-p_v|$*.*

Proof. Every $s\in\mathcal{S}_{\mathrm{ext}}(H)$ satisfies $s(u)-s(v)=0$, while every $s\notin\mathcal{S}_{\mathrm{ext}}(H)$ satisfies $|s(u)-s(v)|\le 1$, so $|p_u-p_v|\le\sum_{s\notin\mathcal{S}_{\mathrm{ext}}(H)}\mu_s|s(u)-s(v)|\le\sum_{s\notin\mathcal{S}_{\mathrm{ext}}(H)}\mu_s$. □

Theorem 24 (Quantum preparations in $\mathcal{P}(H_{17,10}) \setminus \mathcal{P}_{\text{ext}}(H_{17,10})$)**.** *In the real FOR of $H_{17,10}$ (Table S2 of the supplementary material):*

1. *The pure quantum preparation along $r_{24} = (1, -1, 2)^T$,*

$$\rho_\star = \frac{1}{6}\begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{pmatrix}, \qquad p_i = \mathrm{Tr}(\rho_\star P_i) = \frac{(r_{24} \cdot r_i)^2}{6\|r_i\|^2}, \tag{26}$$

belongs to $\mathcal{P}(H_{17,10})$ via the exact five-state convex combination (state indices in Table S40 of the supplementary material; see also Tables S46–S47)

$$p = \frac{1}{12}s_4 + \frac{5}{36}s_9 + \frac{1}{36}s_{10} + \frac{7}{12}s_{14} + \frac{1}{6}s_{17}, \tag{27}$$

yet satisfies $p_{24} = 1$ and $p_{10} = 8/9$ for the chromatically inseparable pair $(10, 24)$, so

$$p \notin \mathcal{P}_{\text{ext}}(H_{17,10}) \qquad \text{and} \qquad \eta_{H_{17,10}}(p) = \frac{1}{9}. \tag{28}$$

2. *The pure preparation along the coordinate ray $z_3 = r_5 = (0, 0, 1)^T$, with Born vector $p'_i = (r_5 \cdot r_i)^2/\|r_i\|^2$, has $p'_{24} = p'_{18} = 2/3$ and $p'_{10} = p'_{11} = 1/3$, belongs to $\mathcal{P}(H_{17,10})$ via the four-state convex combination $p' = \frac{1}{3}s_3 + \frac{1}{6}s_4 + \frac{1}{3}s_{14} + \frac{1}{6}s_{15}$, and satisfies $\eta_{H_{17,10}}(p') = 1/3$.*

3. *For every density operator ρ and both inseparable pairs $(u, v) \in \{(10, 24), (11, 18)\}$, $|p_u(\rho) - p_v(\rho)| \le \|P_u - P_v\| = 1/3$. Hence Proposition 23 certifies at most $1/3$ on $H_{17,10}$, and the preparation in (2) attains this value.*

Proof. (1) Direct evaluation of Eq. (27) on all 17 vertices matches p_i in Eq. (26) (Table S46), with $r_{10} = (1, -1, 1)^T$ giving $p_{10} = 4^2/(6 \cdot 3) = 8/9$. Since $10 \sim_c 24$, Proposition 23 gives $\eta_{H_{17,10}}(p) \ge |1 - 8/9| = 1/9$, which is attained in Eq. (27) because among $\{s_4, s_9, s_{10}, s_{14}, s_{17}\}$ precisely s_4 and s_{10} are nonextendable, with total weight $1/12 + 1/36 = 1/9$. (2) Direct evaluation of the four-state mixture on all 17 vertices reproduces p' (Tables S46–S47). Proposition 23 gives $\eta_{H_{17,10}}(p') \ge |2/3 - 1/3| = 1/3$, attained because s_3, s_{14} are extendable and s_4, s_{15} are not, with total weight $1/6 + 1/6 = 1/3$. (3) For unit vectors with squared overlap κ, $P_u - P_v$ has eigenvalues $\pm\sqrt{1 - \kappa}$ and 0. Both pairs have $\kappa = (r_u \cdot r_v)^2/(\|r_u\|^2\|r_v\|^2) = 16/18 = 8/9$, so $|\,\mathrm{Tr}\,\rho(P_u - P_v)| \le \|P_u - P_v\| = 1/3$. □

The exact support-and-weight certificates and the complete Born probability vectors are given in the supplement. These tables are regenerated from the checked state inventory, rather than manually transcribed.

These preparations exclude only models built from noncontextual eigenvalue assignments: their ordinary-state models are exhibited explicitly. A state is extendable exactly when some noncontextual eigenvalue assignment c makes it the indicator of a single eigenvalue λ_a in every context observable A_e of Eq. (1). For a model built from noncontextual eigenvalue assignments, every chromatic identity $u \sim_c v$ entails the equality $p_u = p_v$; quantum mechanics violates it, and Proposition 23 converts the violation into a lower bound on η. The weight η therefore measures how far the quantum predictions force any noncontextual truth-assignment model away from noncontextual eigenvalues: on $H_{17,10}$, at least one third of the weight in the preparation along z_3. Since $\mathcal{P}_{\text{ext}}(H_{17,10})$ lies in the hyperplane $p_{10} = p_{24}$, a Born vector in $\mathcal{P}(H_{17,10})$ generically lies outside $\mathcal{P}_{\text{ext}}(H_{17,10})$; the content of Theorem 24 is the size of the required weight. Conversely, averaging the d color classes of any admissible coloring gives the maximally mixed Born vector $1/d$, so the restricted obstruction is preparation dependent. The $H_{17,10}$ ordinary implication $s(10) \le s(24)$ also prevents order

determination for distinct ray atoms; vertex separation and order representation must not be conflated.

For a single system, nothing forces a classical model to use only extendable states. The supersinglet game of Sec. I does. A local hidden-variable model that reproduces its perfect correlations gives the d parties the d color classes of a strong d-coloring, so each party follows an extendable state. If $d-1$ parties measure the same basis and obtain all of its rays but v, the last particle is prepared along v, and in such a model its outcome probabilities lie in $\mathcal{P}_{\mathrm{ext}}(H)$. On $H_{17,10}$ the two preparations of Theorem 24 arise in this way from the bases $\{4,11,24\}$ and $\{2,8,5\}$. Both lie outside $\mathcal{P}_{\mathrm{ext}}(H_{17,10})$, so quantum mechanics contradicts every such local model, although $\chi(H_{17,10})=3$. More directly, if $u\sim_c v$, such a model never lets two different parties obtain u and v. Quantum mechanics gives this event the probability $(1-\kappa)/[d(d-1)]$, where κ is the squared overlap of the two rays, because the reduced state of two particles of the supersinglet is the normalized projector onto the antisymmetric subspace. For the pairs $(10,24)$ and $(11,18)$ of $H_{17,10}$ this probability is $1/54$.

B. Order determination versus Born probabilities on H_{37}, $H_{\mathrm{YO}}^{(4)}$, and H_{65}

When $\chi(H)>d$ (Regime I), no noncontextual eigenvalue assignment exists, $\mathcal{P}_{\mathrm{ext}}(H)=\varnothing$, and the relevant probability question is whether Born assignments $p_i(\rho)=\mathrm{Tr}(\rho\Pi_i)$ lie in $\mathcal{P}(H)=\mathrm{conv}\,\mathcal{S}(H)$ (Table IV).

Corollary 25 (Order determination does not imply classical probability models)**.** *The* 76*-event orthomodular lattice $L(H_{37})$ admits an order-determining state space $\mathcal{S}(H_{37})$ and a classical mixture for the maximally mixed state $I_3/3$, yet every $p\in\mathcal{P}(H_{37})$ satisfies the tight inequality*

$$\mathcal{C}_{37}(p)=\sum_{k=0}^{2}\big[p(P_k)+p(Q_k)\big]\le 2, \tag{29}$$

whereas the complex FOR of H_{37} satisfies

$$\sum_{k=0}^{2}(\Pi_{P_k}+\Pi_{Q_k})=\mathrm{diag}(3/2,3/2,3)=\frac{3}{2}I_3+\frac{3}{2}|e_3\rangle\langle e_3|, \tag{30}$$

yielding $\mathcal{C}_{37}(\rho)=\frac{3}{2}(1+\rho_{33})>2$ whenever $\rho_{33}>1/3$ (and $\mathcal{C}_{37}(|e_3\rangle\langle e_3|)=3$).

Proof. In the state parametrization of Theorem 19(4), every $t\in\mathbb{Z}_3^3$ has $|U(t)|+|W(t)|\le 2$, establishing Eq. (29), while summing the normalized projectors of P_k,Q_k in Eq. (19) using $1+\zeta+\zeta^2=0$ gives Eq. (30). Uniform weights $1/6$ on $\mathcal{R}_6$ reproduce $p_i(I_3/3)=1/3$. □

In dimension four, the non-linear suspension $H_{\mathrm{YO}}^{(4)}$ obeys $\mathcal{Y}(p)=p_{26}+\sum_{i=10}^{13}p_i\le 1$ on $\mathcal{P}(H_{\mathrm{YO}}^{(4)})$, whereas $\sum_{i=10}^{13}\Pi_i=\frac{4}{3}(I_4-\Pi_{26})$ gives $\mathcal{Y}(\rho)=1+(1-p_{26}(\rho))/3>1$ for every density operator $\rho\neq\Pi_{26}$[15]. For the canonical linear H_{65}, every $s\in\mathcal{S}(H_{65})$ has 4 active core vertices, so every $p\in\mathcal{P}(H_{65})$ obeys $\sum_{i=1}^{17}p_i=4$. However, with $q=(1,1,1,-1)^T$ and $Q=qq^T/4$, direct summation gives

$$\sum_{i=1}^{17}\Pi_i=4I_4+Q\quad\Longrightarrow\quad\sum_{i=1}^{17}p_i(\rho)=4+\mathrm{Tr}(\rho Q), \tag{31}$$

violating the classical bound 4 whenever $\mathrm{Tr}(\rho Q)>0$ (including $I_4/4$, where the sum is $17/4=\chi_f$, and $\rho=Q$, where it is 5). Nevertheless, on the orthogonal subspace $q^{\perp}$, the density operator $\rho_\star=(I_4-Q)/3$ has core Born probabilities $1/4$ on $C_1\cup C_2$, $1/3$ on $\{9,10,11\}$, and $1/6$ on $B_1\cup B_2\cup B_3$, and completion probabilities $r_{m,1}+r_{m,2}=7/12$. Choosing $\alpha\in\{1,\dots,24\}$ uniformly and $\lambda\in[0,1]$ uniformly and setting $\varepsilon_m=\mathbf{1}_{\{\lambda<\tau_m\}}$ with

TABLE IV. Event order and preparation-dependent probability models for the fixed realizations. Membership statements concern Born vectors $p(\rho)$; the restricted model is $\mathcal{P}_{\mathrm{ext}}$.

Configuration	Order determination	An ordinary-state model	A failed model or witness
$H_{17,10}$	No: $s(10) \leq s(24)$ is a false ray inclusion.	P_{24} and P_{z_3} have ordinary mixtures.	Neither lies in $\mathcal{P}_{\mathrm{ext}}$; $\eta = 1/9$ and $1/3$ (maximal pair certificate).
Completed Yu–Oh	No.	No Born preparation lies in $\mathcal{P}$.	Every preparation gives $\sum_{10}^{13} p_i = 4/3 > 1$.
H_{37}	Yes on its specified 76-event lattice; 18 states minimal.	$I_3/3$ has a six-state mixture.	$\rho_{33} > 1/3$ violates the P, Q bound.
$H_{\mathrm{YO}}^{(4)}$	No.	Pure apex P_{26} has its singleton-state model.	Every $\rho \neq P_{26}$ violates the displayed bound.
H_{65}	No: $s(16) \leq s(9)$ is a false ray inclusion.	$(I_4 - Q)/3$ has a model for all 65 rays.	$\mathrm{Tr}(\rho Q) > 0$ violates the core sum bound.

$\tau_m = \frac{12}{7} r_{m,2} \in \{4/9, 5/11, 6/11, 5/9\}$ on the 14 missed bridges of S_α gives $\mathbb{E}[s(w_{m,2})] = \frac{14}{24}\tau_m = r_{m,2}$, expressing $p(\rho_\star) \in \mathcal{P}(H_{65})$ as a convex mixture of at most 120 states. Rotating V_{core} by $D_0 = I_4, D_1 = \mathrm{diag}(1,1,-1,-1), D_2 = \mathrm{diag}(1,-1,1,-1), D_3 = \mathrm{diag}(1,-1,-1,1)$ gives a fixed enlarged family with

$$\sum_{a,i} \Pi_i^{(a)} = 17 I_4, \qquad \sum_{a,i} p_i^{(a)} \leq 16 \text{ classically.} \tag{32}$$

Each copy admits at most four true rays under geometric pair exclusions; repeated projectors are counted with multiplicity. The 24 distinct rays form the Peres set[35]. This is an enlargement, not a state-independent test of the original 65 rays.

VII. KS ANCESTRY, SPECIFIED PARENTS, AND ABSTRACT VERSUS GEOMETRIC STATES

A. The KS ancestry hypothesis and abstract transverse extensions

Motivated by the recurring parent structures of our examples, we formulate the *KS ancestry hypothesis*: informally, every noncolorable ($\chi(H) > d$) configuration has a larger KS ancestor. For realized ray configurations, its existence form follows from an elementary union argument, even without the chromatic premise; the substantive open question is whether the retained rays can be made essential to the contradiction (Sec. VIII):

Remark 26 (Same-dimensional finite KS ancestry). Every finite ray set $R \subset \mathbb{P}(\mathbb{F}^d)$ ($\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $d \geq 3$) is a proper subset of a finite KS ray set $P \subset \mathbb{P}(\mathbb{F}^d)$ whose orthogonality graph and full-basis hypergraph induce those on R, with every ray of P in a complete context. Consequently, a finite graph G has an injective edge-faithful realization in $\mathbb{F}^d$ if and only if G is an induced subgraph of a finite KS orthogonality graph in $\mathbb{F}^d$.

Indeed, choose a finite full-basis KS set $K \subset \mathbb{P}(\mathbb{F}^d)$[4,36], orient K by $U \in \mathrm{U}(d)$ (or $\mathrm{O}(d)$) so $UK \not\subseteq R$, and complete any uncovered rays of $R \cup UK$ to bases to obtain P. Any state on P would restrict to a state on UK, a contradiction, and keeping the rays of R fixed preserves all induced orthogonalities and full bases on R. The contradiction uses only the added set UK; the rays of R play no role in it.

By Theorem 11, H_8 has no injective realization, and H_9, G_{32} have no ray assignment; none can embed into any 3-dimensional geometric KS parent. For abstract hypergraphs, Lemma 3 yields a direct chromatic-to-stateless construction:

Proposition 27 (Transverse chromatic-to-stateless extension)**.** *For any finite d-uniform hypergraph $H = (V, E)$ ($d \geq 3$), let $T(H)$ take $d-1$ disjoint copies $H^{(1)}, \ldots, H^{(d-1)}$ and adjoin for each $v \in V$ a fresh vertex w_v and the transverse context $T_v = \{v^{(1)}, \ldots, v^{(d-1)}, w_v\}$. Then $T(H)$ has $d|V|$ vertices, $(d-1)|E| + |V|$ contexts, is linear if H is linear, and its two-valued states correspond bijectively to labeled strong d-colorings of H; in particular,*

$$\mathcal{S}(T(H)) = \varnothing \quad \Longleftrightarrow \quad \chi(H) > d. \tag{33}$$

Proof. A state $s \in \mathcal{S}(T(H))$ restricts to $d-1$ states on $H^{(k)}$ that are pairwise disjoint by T_v (forcing $s(w_v) = 1 - \sum_{k=1}^{d-1} s(v^{(k)})$), which by Lemma 3 is equivalent to a strong d-coloring of H. □

In dimension $d = 3$, any edge $u \sim v$ of H creates a Berge 4-loop $(u^{(1)}, v^{(1)}, v^{(2)}, u^{(2)})$ in $T(H)$, excluding an injective realization in $\mathbb{C}^3$ by the common orthogonal-complement argument; thus $T(H)$ is an abstract construction for a single system. It describes, however, the supersinglet game of Ref.[8] (Sec. I): the copies $H^{(k)}$ are the bases measured by $d-1$ of the parties, the transverse context T_v states that exactly one party obtains v, and the fresh vertex w_v stands for the last party, whose truth assignment is fixed by Lemma 3. Equation (33) is thus the combinatorial content of the statement that this game has no local hidden-variable model exactly when $\chi(H) > d$. For H_9, $T(H_9)$ (45 vertices, 33 contexts) is context-critical for statelessness on all 33 contexts; for H^+_{YO}, $T(H^+_{\mathrm{YO}})$ (75 vertices, 57 contexts) is critical on all 32 horizontal contexts and 12 transverse contexts $v \in \{14, \ldots, 25\}$, but remains stateless upon deleting T_v for $v \in \{1, \ldots, 13\}$ (Sec. S3 of the supplementary material).

B. Explicit inclusions in known KS parents and Cabello completions

Beyond unrestricted existence, the following explicit unions use known KS components; the unions are not asserted to be previously named, smallest, or critical parents (explicit coordinate dictionaries and certificates in Secs. S1–S3 of the supplementary material):

1. *Bub's 49-ray context-critical real parent of H^+_{YO}:*

   Proposition 28 (Context-critical real parent of H^+_{YO})**.** *Adjoining to H^+_{YO} the eight rays $(0, 1, \pm 2)$, $(0, 2, \pm 1)$, $(1, \pm 2, 0)$, $(2, \pm 1, 0)$ yields 33 rays with 20 bases and 16 uncovered orthogonal pairs that eliminate all 24 states of H^+_{YO}. Completing those 16 pairs yields the 49-ray, 36-context full-context Schütte–Bub KS configuration[20,37], containing H^+_{YO} as an induced restriction, in which* every *one of the 36 contexts is critical for $\mathcal{S}_{\mathrm{ctx}} = \varnothing$ (Table S83 of the supplementary material). If the orthogonalities of a deleted context are retained as geometric exclusions, only the 20 bases of the 33-ray stage remain critical; deleting any of the 16 completion contexts leaves $\mathcal{S}_{\mathrm{geom}} = \varnothing$.*

2. *MUB-inventory parents of H_{37}:* In the 165-ray, 130-context MUB inventory[18] formed by three 69-ray, 50-context KS lineages L_0, L_1, L_2 (where each L_a completes the 36 uncovered pairs of Cabello's 33-ray, 14-basis KS set[38]; Pavičić[39] points out that this 33-ray set already occurs as a subset of a 69-ray, 50-basis set in Ref.[40]), H_{37} embeds as an induced restriction via the seed-index matrix $j(a,k) = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 9 & 8 \end{smallmatrix}\right)$ mapping $z \leftrightarrow a_{1,1}$, $M_{ak} \leftrightarrow u_{j(a,k)}$, $P_k \leftrightarrow c_{1,k+1}$, $Q_k \leftrightarrow c_{2,k+1}$, $A_k \leftrightarrow b_{1,k+1}$, $B_{ak} \leftrightarrow d_{1,j(a,k)}$, $C_{ak} \leftrightarrow d_{2,j(a,a+k)}$. It lies in the 57-ray, 40-context intermediate spoke hypergraph (297 states) and in each 87-ray, 64-context KS union $H_{37} \cup L_a$.

3. *Four-dimensional Peres and Cabello parents of $H^{(4)}_{\mathrm{YO}}$ and H_{65}:* Taking the union with Peres's 24-ray set P_{24}[35] gives a 40-ray, 36-context KS parent of $H^{(4)}_{\mathrm{YO}}$ ($|H^{(4)}_{\mathrm{YO}} \cap P_{24}| = 10$; note that a single-apex lift of a 3-dimensional KS set always retains the unique apex-one state) and a 72-ray, 48-context KS parent of H_{65} ($|H_{65} \cap P_{24}| = 17$). More economically:

Proposition 29 (Three-ray KS extensions of H_{65}). *Adjoining* $a = (1,0,0,1), b = (0,1,0,1), c = (0,0,1,1)$ *to* H_{65} *yields a* 68*-ray,* 35*-basis KS parent containing* H_{65} *as an induced restriction and Cabello's* 18*-ray,* 9*-context parity proof* $C_{18,9}$[21] *(*C_3, C_4, C_5 *plus* $D_1 = \{2,3,12,a\}, D_2 = \{1,3,15,b\}, D_3 = \{1,2,17,c\}, D_4 = \{6,7,13,a\}, D_5 = \{6,8,14,b\}, D_6 = \{7,8,16,c\}$*). Adjoining* $(0,1,-1,0)$, $(1,0,-1,0)$, $(1,-1,0,0)$ *instead yields a* 68*-ray,* 35*-basis KS parent containing a ray-critical* 20*-ray,* 11*-context parity proof* $C_{20,11}$ *(*$11 = 4s(4)+4s(5)+2\sum_{i\neq 4,5} s(i)$*). All* 20 *rays of* $C_{20,11}$ *belong to Peres's* 24*-ray set, and* $C_{20,11}$ *is one of its* 96 *critical* 20*-ray KS subsets, which are mutually isomorphic and include Kernaghan's set*[21,22].

Proof. In $C_{18,9}$, summing s over C_3, C_4, C_5 gives $3 = 2(s(9)+s(10)+s(11))+\sum_{k=12}^{17} s(k)$, so $s(12)+\cdots+s(17)$ is odd, whereas $D_1 \oplus D_4, D_2 \oplus D_5, D_3 \oplus D_6$ give $s(12)+s(13) \equiv s(2)+s(3)+s(6)+s(7)$, $s(14)+s(15) \equiv s(1)+s(3)+s(6)+s(8)$, $s(16)+s(17) \equiv s(1)+s(2)+s(7)+s(8)$ (mod 2), summing to 0 (mod 2). In $C_{20,11}$ (comprising $C_1, \ldots, C_5$ and six contexts on $\{1,\ldots,8,12,\ldots,17\}$ with $(0,1,-1,0)$, $(1,0,-1,0)$, $(1,-1,0,0)$), vertices 4, 5 have degree 4 and the other 18 vertices have degree 2, so $11 = 4s(4)+4s(5)+2\sum_{i\neq 4,5} s(i)$, an odd-equals-even contradiction; explicit states after deleting any single ray are certified in Sec. S3 of the supplementary material. □

Both 68-ray presentations have 35 bases covering 192 of 195 geometric orthogonal pairs and context intersections of size up to two. They are not edge-faithful in the listed bases. The 72-ray Peres union covers all pairs; no size minimum or criticality of either full parent is asserted.

4. *Faithful completion* $C_{36,18}$ *and abstract versus geometric states on* $C_{45,21}$*:*

Corollary 30 (Faithful Cabello completion and vertex deletion). *The* 9 *uncovered orthogonal pairs of* $C_{18,9}$ *satisfy Eq.* (22)*, so Lemma 21 yields a linear edge-faithful* 36*-ray,* 18*-context KS hypergraph* $C_{36,18}$ *in* $\mathbb{R}^4$ *with* $\omega = 4, \chi = 5$*, and* $\mathcal{S} = \varnothing$*. Deleting one vertex (say* c*) and its two contexts from* $C_{18,9}$ *leaves* 17 *rays,* 7 *bases,* 14 *uncovered pairs, and* 12 *core states whose abstract completion* $C_{45,21}$ *has* $\chi = 5$ *and* $|\mathcal{S}_{\mathrm{ctx}}(C_{45,21})| = 3{,}584$ *unital separating context states (*24 *canonical). However,* $v_{16} \in \mathrm{span}(v_1, v_2)$ *and* $v_{17} \in \mathrm{span}(v_7, v_8)$ *violate Eq.* (22)*, forcing four extra orthogonalities that reduce the geometric state space to* $|\mathcal{S}_{\mathrm{geom}}(C_{45,21})| = 2{,}048$ *with* $s(9) = s(16) = s(17) = 0$*, destroying unitality and separation (Table V).*

Proof. For $C_{18,9}$, the 9 uncovered pairs are the 6 core bridges $\{1,13\}$, $\{2,14\}$, $\{3,16\}$, $\{6,17\}$, $\{7,15\}$, $\{8,12\}$ from H_{65} plus $\{11,a\}$, $\{10,b\}$, $\{9,c\}$. They satisfy Eq. (22) and are completed in Tables S28–S29 of the supplementary material. Deleting c, D_3, D_6 leaves 12 core supports: 10 miss 8 of the 14 uncovered pairs and 2 miss 9 (Table S84), giving $10 \cdot 2^8 + 2 \cdot 2^9 = 3{,}584$ context states. Since $v_{16} = v_1 + v_2$ and $v_{17} = \frac{1}{2}(v_7 - v_8)$, completing $\{1,2\}$ by $\{21,22\}$ and $\{7,8\}$ by $\{35,36\}$ forces $v_{16} \perp v_{21}, v_{22}$ and $v_{17} \perp v_{35}, v_{36}$ in the local labels of Tables S31–S32. If $s(16) = 1$, neither 1 nor 2 is selected in any admissible core support, so $s(21)+s(22) = 1$, contradicting geometric exclusion; the same argument gives $s(17) = 0$. The retained hub bases then force $s(9) = 0$. Exactly 8 core supports satisfy $s(9) = s(16) = s(17) = 0$, each missing 8 uncovered pairs. All their fresh-pair choices obey the four extra exclusions, yielding $8 \cdot 2^8 = 2{,}048$ geometric states. □

VIII. DEPENDENCIES, COUNTEREXAMPLES, AND REMAINING QUESTIONS

The common dependency structure is incidence → states and colorings, followed by independent tests of geometry, event order, and quantum probabilities. A chromatic argument can be made on an abstract hypergraph; its operational interpretation requires

TABLE V. Context and geometric state spaces in the four-dimensional examples. Linearity of the listed incidence structure does not imply edge-faithfulness.

Configuration	$\lvert V\rvert/\lvert E\rvert$	$\lvert\mathcal{S}_{\mathrm{ctx}}\rvert$	$\lvert\mathcal{S}_{\mathrm{geom}}\rvert$	Scope
$C_{18,9}$	18/9	0	0	Nine uncovered geometric pairs; parity KS proof.
$C_{36,18}$	36/18	0	0	Faithful linear completion.
$C_{45,21}$	45/21	3,584	2,048	Four additional pairs destroy geometric unitality and separation.
H_{65}	65/29	393,216	393,216	Edge-faithful; all context states respect the geometry.

a ray realization. Conversely, a realization does not select an ontic state family or a classical probability mixture. The exact-palette assumption, that the outcomes admit a noncontextual assignment of d eigenvalues, adds a precise completion test at the level of whole truth assignments: a truth assignment is compatible with noncontextual eigenvalues exactly when the outcomes it leaves false can carry the remaining $d-1$ eigenvalues.

As a worked example, consider how many hidden states a classical model needs for the 13 Yu–Oh rays of Sec. III C. A hidden state is the complete classical description of the system: it fixes, or at least gives the probabilities of, the outcome of every measurement. A preparation produces a hidden state, possibly at random from several, and a quantum state then corresponds to a probability distribution over hidden states. In a generalized urn model[13] the hidden state is the ball that is drawn; in a deterministic noncontextual model it is a two-valued state. One fact suffices: if two rays u and v are orthogonal, a measurement in a basis containing both tells the preparations along u and along v apart with certainty. Hence no hidden state can arise from both preparations, since in that hidden state the measurement would have to yield u with certainty and also v with certainty.

Suppose now that a classical model uses only three hidden states A, B, C, as many as the dimension.

1. The rays $z_1 = (1,0,0)$, $z_2 = (0,1,0)$, $z_3 = (0,0,1)$ are mutually orthogonal, so they need three different hidden states, say A, B, C.

2. The rays $y_1^{\pm} = (0,1,\pm1)$ are orthogonal to z_1 and to each other, so they take B and C in some order. Likewise $y_2^{\pm} = (1,0,\pm1)$ take A and C, and $y_3^{\pm} = (1,\pm1,0)$ take A and B.

3. Each diagonal ray is orthogonal to exactly one ray of each pair: $h_0 = (1,1,1)$ to y_1^-, y_2^-, y_3^-; $h_1 = (1,-1,-1)$ to y_1^-, y_2^+, y_3^+; $h_2 = (1,-1,1)$ to y_1^+, y_2^-, y_3^+; and $h_3 = (1,1,-1)$ to y_1^+, y_2^+, y_3^-. Each diagonal ray needs a hidden state that none of its three neighbors uses.

4. A diagonal ray can use A only if its y_2-neighbor has C and its y_3-neighbor has B. This fixes two of the three signs of its neighbors, and exactly one diagonal ray has the required signs, because the four diagonal rays have the four sign patterns with an even number of plus signs. The same holds for B and for C. So at most three diagonal rays find a free hidden state; the fourth is orthogonal to rays in A, B, and C.

Any classical model therefore needs at least four hidden states merely to tell these 13 preparations apart, whereas a qutrit realizes them in three dimensions. A model with exactly three hidden states, each tied to one eigenvalue, is a noncontextual eigenvalue assignment: in hidden state A, every context yields the outcome that carries λ_0, and likewise for B and C. Steps 1–4 thus also show directly that the Yu–Oh rays admit no noncontextual eigenvalue assignment; Theorem 13 refines this count for completed Yu–Oh.

Table VI also explains why the probability and gadget examples carry different force. A one-way ordinary TITS already gives a symmetric color identity; a two-way purely chromatic identity requires counterstates in both truth directions. A ternary color exclusion may

TABLE VI. Logical boundaries of the integrated results. Each row prevents transfer of a conclusion between different assignment or representation problems.

Tempting unrestricted inference	Counterexample or necessary qualification
$\chi > d$ excludes states or ray realizations	Completed Yu–Oh has faithful real rays and 24 unital separating states.
$\chi > d$ makes I_d/d contextual[8]	One system needs $\chi_f > d$[16]: H_{37} has $\chi = 4$, $\chi_f = 3$, and a six-state mixture for $I_3/3$.
Ordinary states separate ⇒ colors separate	Faithful $H_{17,10}$ and $H_{21,13}$ have forced color identities.
Colors separate ⇒ every state extends	The pentagon separates all vertex pairs but has one nonextendable state.
Vertex separation ⇒ event-order determination	$H_{17,10}$ has 10 ⇒ 24; H_{65} has 16 ⇒ 9 and ray–plane collapse.
Order determination ⇒ all Born mixtures	The specified event order of H_{37} is represented, while $\rho_{33} > 1/3$ violates the state inequality.
$\chi_f > d$ establishes fixed-measurement state independence	Ref.[15] (Thm. 1) (Yu–Oh plus apex); linear and apex-free: H_{65} admits the Born model of $(I_4 - Q)/3$ on all 65 rays.
Abstract stateless extension ⇒ geometric KS parent	Closing a color-identical pair changes orthogonality; transverse four-loops exclude injective triadic geometry.
Generic KS ancestry explains the original chromatic gap	Unrestricted extension holds even for colorable sets and need not use the old rays essentially.

survive even when every pair can be equal. A strict inclusion $\mathcal{P}_{\text{ext}} \subsetneq \mathcal{P}$ therefore need not be detectable through pair equalities alone.

Open problems concern global minimum faithful examples beyond the specified inventory, economical rational completions with fewer degree-one rays, residual-coloring certificates in dimensions above three, and geometric ancestry in which the retained rays are essential to the contradiction. The current statements establish existence and the recorded finite classifications, without asserting these stronger minima or essential-ancestry properties. All complete enumerations and the detailed verification scope remain in the supplement and electronic records.

IX. CONCLUSION

We started from a simple physical requirement on intertwined complete experiments: that each is the measurement of a maximal, nondegenerate observable with one common spectrum of d eigenvalues, and that the eigenvalue carried by an outcome is an element of physical reality in the sense of Einstein, Podolsky, and Rosen[3], the same in every experiment in which the outcome occurs. This requirement of noncontextual eigenvalues is exactly strong d-colorability of the context hypergraph. It asks for more than Kochen–Specker noncontextuality: not one classical truth assignment, but d of them with disjoint supports covering every outcome. Its failure, $\chi(H) > d$, is therefore a weaker but more widespread form of contextuality. Completed Yu–Oh has 24 truth assignments separating all of its rays, yet its outcomes cannot carry three eigenvalues noncontextually. These truth assignments also give a classical realization, which fails in the same way, so $\chi(H) > d$ restricts the spectra of maximal observables rather than truth values. When H is realized in $\mathbb{C}^d$, it also means that a classical model needs more than d hidden states merely to tell apart preparations along orthogonal rays.

By unifying these common-spectrum chromatic obstructions ($\chi(H) > d$) with exact-palette chromatic separability constraints ($\chi(H) = d$, $\mathcal{S}_{\text{ext}}(H) \subsetneq \mathcal{S}(H)$), we have distinguished the requirements across noncontextual eigenvalue assignments, geometric ray realizations, orthomodular event representations, and quantum probability models. At the foundation,

$\chi(H) > d$ excludes noncontextual eigenvalues for uniform maximal context observables, whereas the exclusion of an injective realization for H_8 and of every ray assignment for H_9, G_{32} requires direct Hermitian projection forcing, correcting the earlier spectral argument for G_{32}[6] and establishing four-quadrant independence even at Berge girth five. Within chromatically complete hypergraphs ($\chi(H) = d$), where noncontextual eigenvalue assignments exist, our completion criterion ($\chi(\Gamma(H)[V \setminus \operatorname{supp} s]) \leq d - 1$) characterizes which truth assignments arise from noncontextual eigenvalue assignments and explains the specified examples of chromatic identities ($u \sim_c v$) and purely chromatic TIFS/TITS/ternary constraints ($H_{17,10}, H_{21,13}, D_{16,9}, T_{19,11}, W^{(4)}_{16,6}$) and yields measurable nonextendable weights 1/9 and 1/3 for quantum ray preparations on $H_{17,10}$, the latter being the largest value a single chromatic identity can certify there. In higher complexity, the Hesse hypergraph H_{37} (6/7/18 OML minima) and our rational four-dimensional completion H_{65} (393,216 states) demonstrate that neither exact order determination nor a fractional chromatic gap decides classicality for all Born preparations, while the elementary KS ancestry result, the 68-ray KS parents of H_{65}, and the completions $C_{36,18}$ and $C_{45,21}$ of $C_{18,9}$ delineate unrestricted embedding from specified critical parents and geometric pair exclusions.

The common thread is that a classical account of quantum outcomes can succeed at the level of truth values and still fail at the level of eigenvalues. For the preparation along z_3, the quantum predictions on $H_{17,10}$ admit noncontextual truth-assignment models, but every such model must place at least one third of its weight on truth assignments that no noncontextual eigenvalue assignment produces.

SUPPLEMENTARY MATERIAL

See the supplementary material for the explicit vector and context inventories, the complete two-valued state and coloring enumerations, the odd-cycle and separation-profile certificates, and the supplementary proofs and verification reductions.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its supplementary material. The verification programs and machine-readable certificates are available from the author upon reasonable request.

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